

Introduction to Numerical Software

Presented to
ATPESC 2025 Participants

Ulrike Meier Yang
Lawrence Livermore National Laboratory

Date 08/05/2025



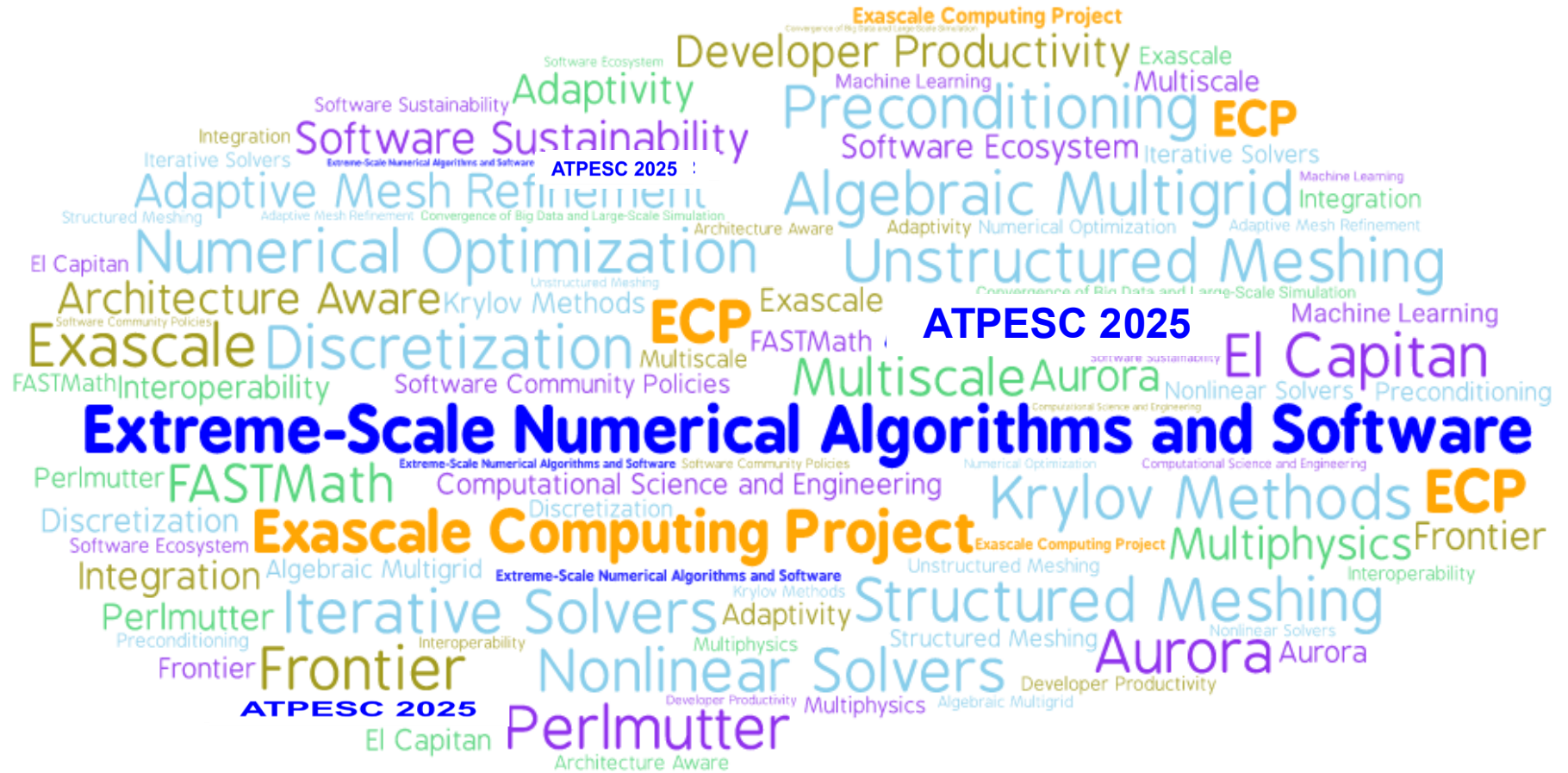
ATPESC Numerical Software Track

ATPESC 2025
EXTREME-SCALE COMPUTING



Outline

- Logistics for the day
- Intro to numerical algorithms and software for extreme-scale science

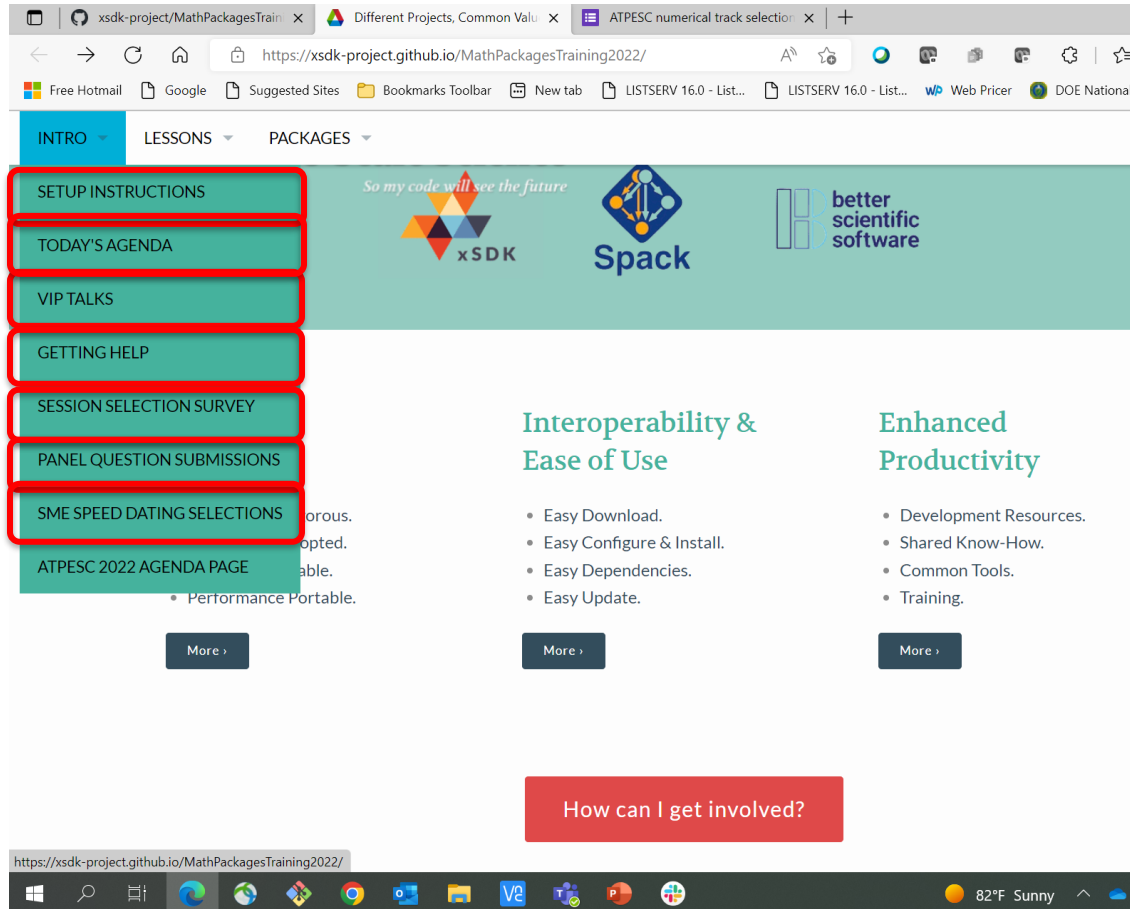


Your home bases for the day: ATPESC Track 5

Numerical Algorithms and Software for Extreme-Scale Science

- Main ATPESC Agenda
 - <https://extremecomputingtraining.anl.gov/agenda-2025/#Track-5>
 - slides (pdf) and presenter bios
- Math Packages Training Site
 - session abstracts, links to parallel breakout rooms, hands-on lessons, more
 - <https://xsdk-project.github.io/MathPackagesTraining2025/agenda/>

<https://xsdk-project.github.io/MathPackagesTraining2025/>



- Setup instructions
- Today's agenda
- VIP talks
- Getting help
- Session Selection Survey
- Panel question submission
- SME speed dating selections

Agenda

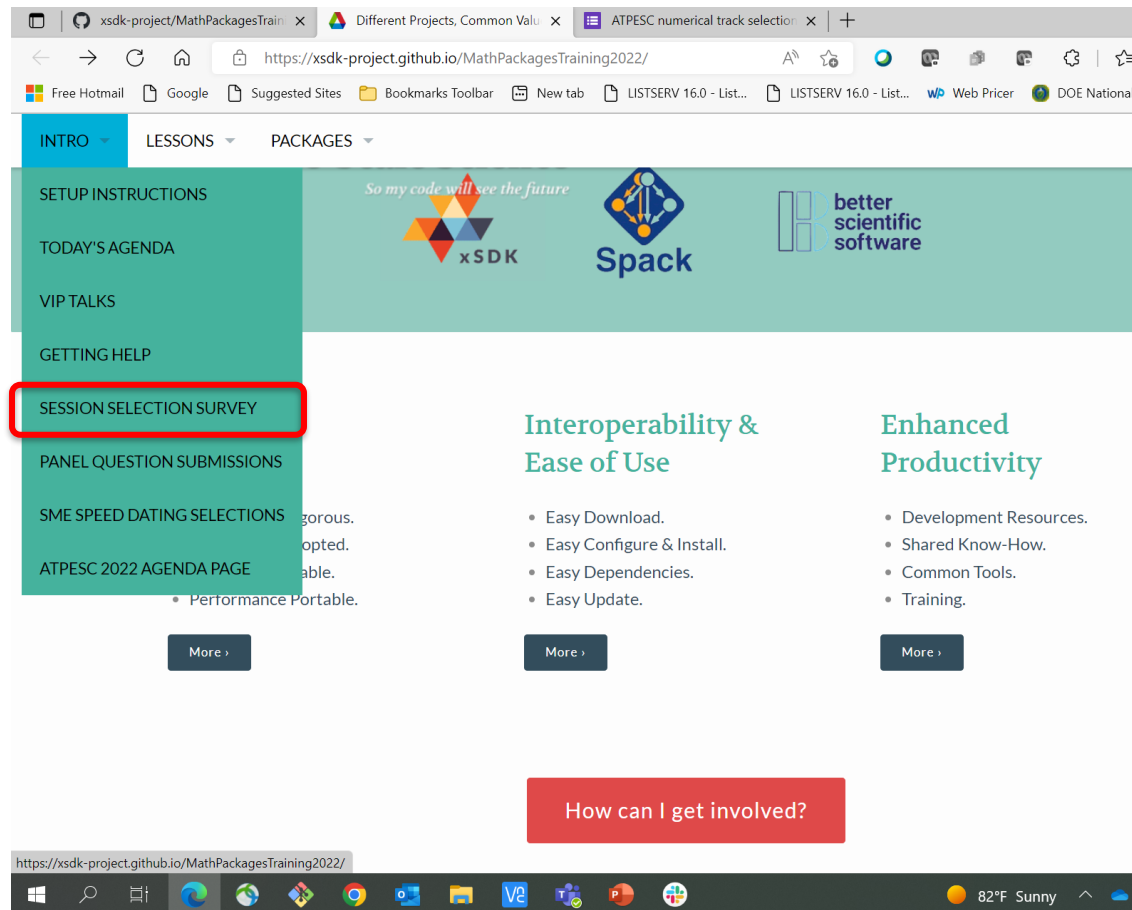
<https://extremecomputingtraining.anl.gov/agenda-2025/#Track-5>

Time	Room?	Room?
8:30 – 9:30	Introduction to Numerical Software – Ulrike Yang	
9:30 – 10:45	Structured Discretization (AMReX) – Andrew Myers, Weiqun Zhang	Unstructured Discretization (MFEM/PUMI) – Mark Shephard, Cameron Smith, Mark Stowell
10:45 – 11:15	Break, Subject Matter Expert (SME) Selections, Panel Questions	
11:15 – 12:30	Iterative Solvers & Algebraic Multigrid (hypre) – Daniel Osei-Kuffuor, Ulrike Yang	Direct Solvers (SuperLU, STRUMPACK) – Sherry Li, Yang Liu
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1:30 – 2:45	Nonlinear Solvers (PETSc) – Richard Mills	Time Integration (SUNDIALS) – David Gardner
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5:15 – 6:30	Optional Activity: SME Speed-dating	
6:30 – 7:30	Dinner	
7:30 – 9:00	After-Dinner Talk: Jim Demmel	

Choose which lecture you want to attend!

Access: [ATPESC numerical track selections](#)

<https://xsdk-project.github.io/MathPackagesTraining2025/>



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Free Hotmail Google Suggested Sites Bookmarks Toolbar New tab LISTSERV 16.0 - Lis

ATPESC numerical track selections

Participants, please denote sessions you plan to attend so that needed room size can be determined

yang11@llnl.gov [Switch account](#) ☁

* Indicates required question

Email *

Your email

Parallel session One *

☐ Structured Meshes (with AMReX)

☐ Unstructured Meshes (with MFEM/PUMI)

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Block-structured adaptive mesh refinement framework. Scalable support for hierarchical mesh and particle data, with embedded boundaries.

Capabilities

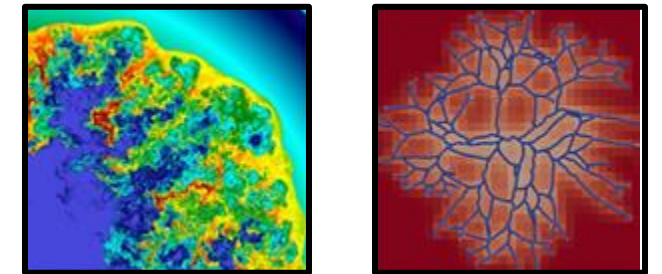
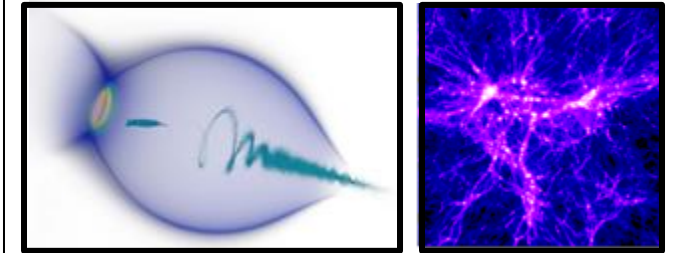
- Support for PDEs on a hierarchical adaptive mesh with particles and embedded boundary representations of complex geometry
- Support for multiple modes of time integration
- Support for explicit and implicit single-level and multilevel mesh operations, multilevel synchronization, particle, particle-mesh and particle-particle operations
- Hierarchical parallelism –
 - hybrid MPI + OpenMP with logical tiling on multicore architectures
 - hybrid MPI + GPU support for hybrid CPU/GPU systems (NVIDIA CUDA, AMD HIP, Intel SYCL)
- Native multilevel geometric multigrid solvers for cell-centered and nodal data
- Highly efficient parallel I/O for checkpoint/restart and for visualization – native format supported by Visit, Paraview, yt

Open source software

- Used for diverse apps, including accelerator modeling, astrophysics, combustion, cosmology, multiphase flow, phase field modeling, atmospheric modeling and more
- Source code and development hosted on github with rigorous testing framework
- Extensive documentation, examples and tutorials



Examples of AMReX applications

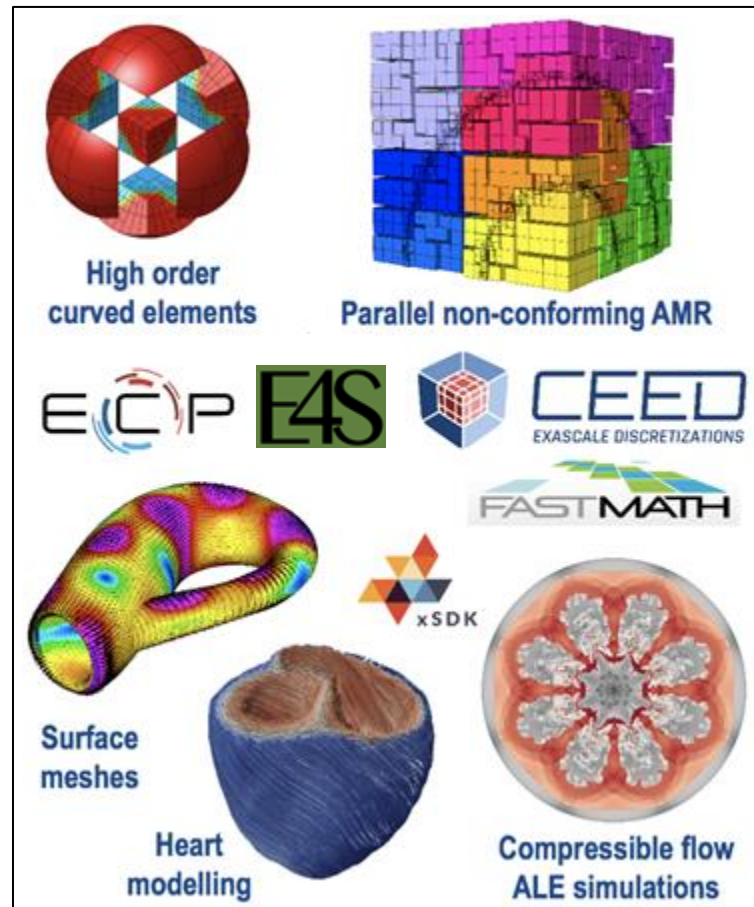


<https://www.github.com/AMReX-Codes/amrex>



Free, lightweight, scalable C++ library for finite element methods. Supports arbitrary high order discretizations and meshes for wide variety of applications.

- **Flexible discretizations on unstructured grids**
 - Triangular, quadrilateral, tetrahedral and hexahedral meshes.
 - Local conforming and non-conforming refinement.
 - Bilinear/linear forms for variety of methods: Galerkin, DG, DPG, ...
- **High-order and scalable**
 - Arbitrary-order H^1 , $H(\text{curl})$, $H(\text{div})$ - and L^2 elements. Arbitrary order curvilinear meshes.
 - MPI scalable to millions of cores and includes initial GPU implementation. Enables application development on wide variety of platforms: from laptops to exascale machines.
- **Built-in solvers and visualization**
 - Integrated with: HYPRE, SUNDIALS, PETSc, SUPERLU, ...
 - Accurate and flexible visualization with VisIt and GLVis
- **Open source software**
 - BSD with thousands of downloads/year worldwide.
 - Available on GitHub, also via OpenHPC, Spack. Part of ECP's CEED co-design center.



<https://mfem.org>

Parallel Unstructured Mesh Infrastructure

Parallel management and adaptation of unstructured meshes.
Interoperable components to support the
development of unstructured mesh simulation workflows

■ Core functionality

- Distributed, conformant mesh with entity migration, remote read only copies, fields and their operations
- Link to the geometry and attributes
- Mesh adaptation (straight and curved), mesh motion
- Multi-criteria partition improvement
- Distributed mesh support for Particle In Cell methods

■ Designed for integration into existing codes

- xSDK package; installs with Slack
- Permissive license enables integration with open and closed-source codes

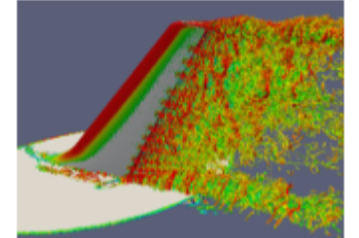
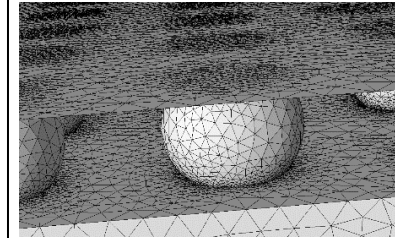
■ In-memory integrations developed

- MFEM: High order FE framework
- PetraM: Adaptive RF fusion
- PHASTA: FE for turbulent flows
- FUN3D: FV CFD
- Proteus: Multiphase FE
- ACE3P: High order FE for EM
- M3D-C1: FE based MHD
- Nektar++: High order FE for flow
- Albany/Trilinos: Multi-physics FE

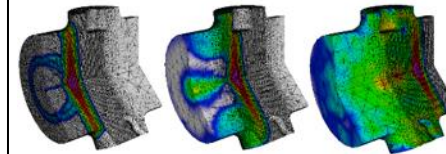
PUMI



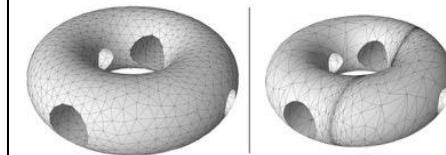
Rensselaer



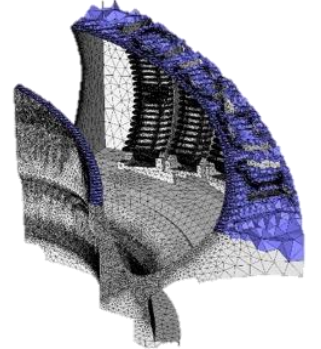
Applications with billions of elements: flip-chip (L), flow control (R)



Mesh adaptation for evolving features



Anisotropic adaptation for curved meshes



RF antenna and plasma surface in vessel.

Source Code: github.com/SCOREC/core
User Guide: scorec.rpi.edu/pumi/PUMI.pdf
Paper: scorec.rpi.edu/REPORTS/2014-9.pdf

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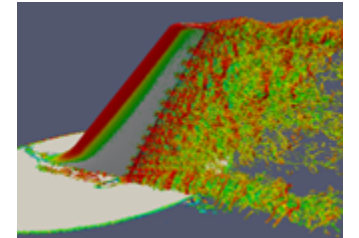
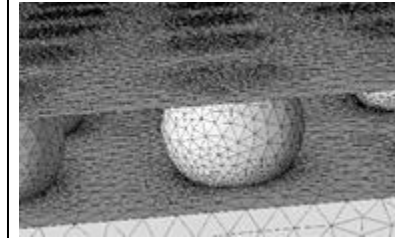
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PUMi

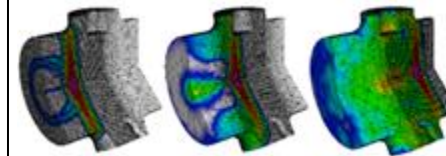


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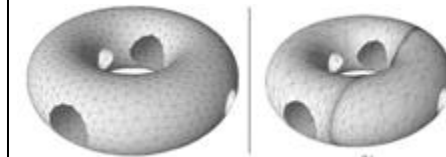
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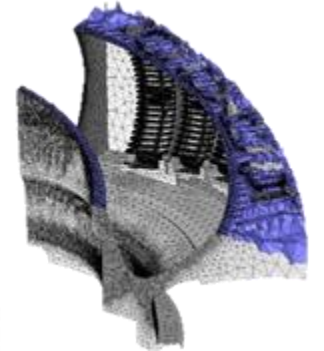
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User Guide:

doi.org/10.5281/zenodo.16412599

Paper: doi.org/10.1145/2814935

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Omega_h Parallel Unstructured Mesh Adaptation on GPUs

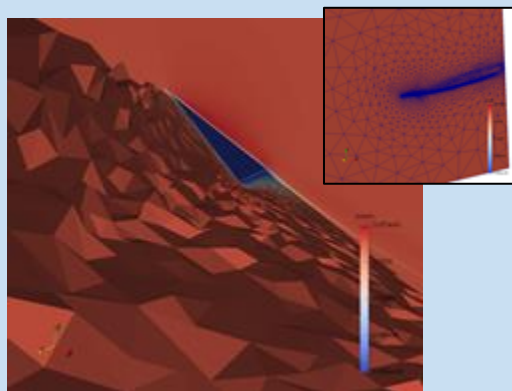
Parallel adaptation of unstructured meshes on GPUs.
Support the development of unstructured mesh simulation workflows on leadership systems.

■ Core functionality

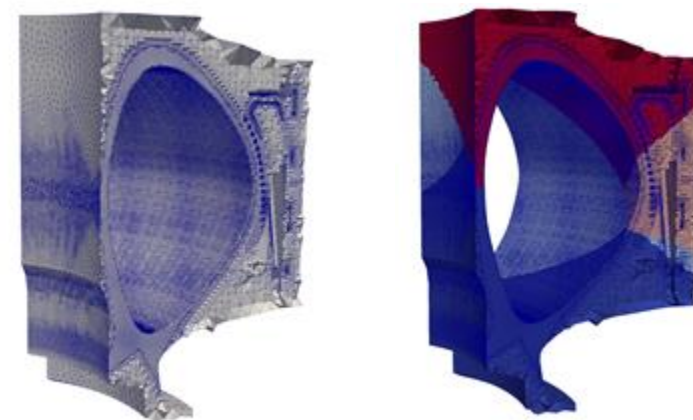
- Distributed, conformant mesh adaptation (coarsening past initial mesh + refinement)
- Manycore and GPU parallelism using Kokkos
- Runs on NVIDIA, AMD, and Intel GPUs
- Supports complex geometric models via Gmsh and Simmetrix SimModSuite

■ Applications Supported

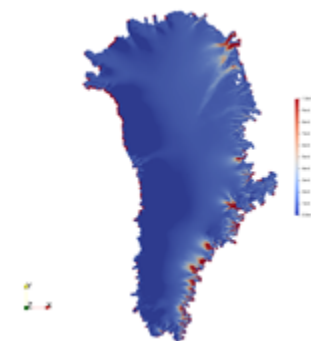
- GITRm: impurity transport
- XGCm: core+edge fusion plasma physics
- MFEM: finite element framework specializing in high-order methods
- PetraM: RF Fusion
- MALI: land ice melting



'Crinkle clip' view of wing's top surface after adaptation (main) and wake on symmetry surface (inset).



Serial and RIB partitioned mesh of RF antenna and vessel model.



Omega_h mesh of the Greenland ice sheet, colored by velocity.

Source Code: github.com/SCOREC/omega_h
Thesis: hdl.handle.net/20.500.13015/1817

PUMIPic Parallel Unstructured Mesh Infrastructure for Particle-in-Cell

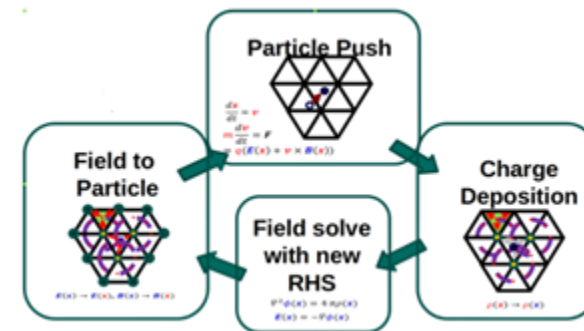
Parallel management of unstructured meshes with particles.
Framework for GPU accelerated particle-in-cell applications using unstructured meshes.

Core functionality

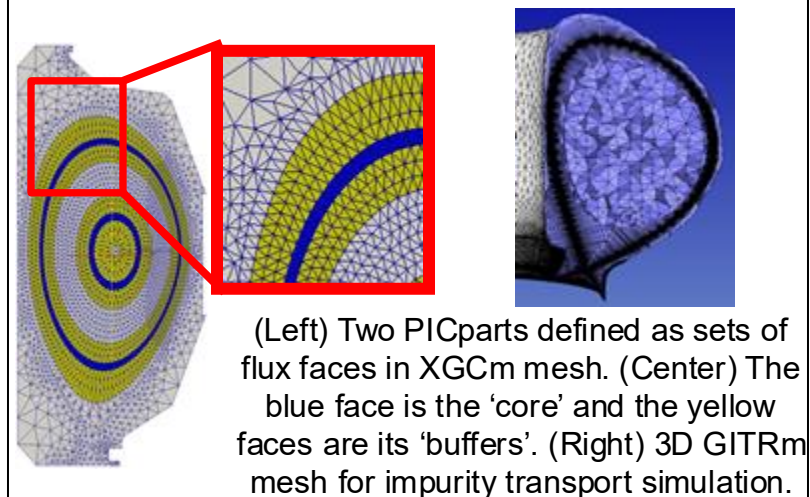
- Unstructured mesh-based approach
 - Particles accessed through mesh
 - Particle search through mesh adjacencies
 - Effective coupling to PDE solvers
 - Partitioning using bounding flux surfaces, graph, or geometric methods
 - PICpart: owned elements + copied elements from topologically or spatially neighboring processes
 - Stored on GPU using Omega_h library: github.com/SNLComputation/omega_h
- Particles
 - Supports multiple species
 - Particle storage choices: Sell-C-Sigma [Kreutzer 2014], COPA Cabana, and DPS.
 - DPS storage optimized for applications with tens of particles per mesh element.
- Parallel kernel launch function abstracts underlying particle and mesh storage
- Supports NVIDIA and AMD GPUs

Applications Supported

- GITRm: impurity transport
- XGCm: core+edge fusion plasma physics
- PolyMPO: GPU polygonal mesh-based material point operations
- PUMI-Tally: supports Monte Carlo neutral particle transport
- Weak scaling on up to 24,000 GPUs of Summit with 1.15 trillion particles running push, particle-to-mesh, and mesh-to-particle operations with an XGCm tokamak mesh and domain decomposition



Stages of a PIC application supported by PUMIPic



Source Code: github.com/SCOREC/pumi-pic
Paper: doi.org/10.1016/j.jpdc.2021.06.004

Agenda

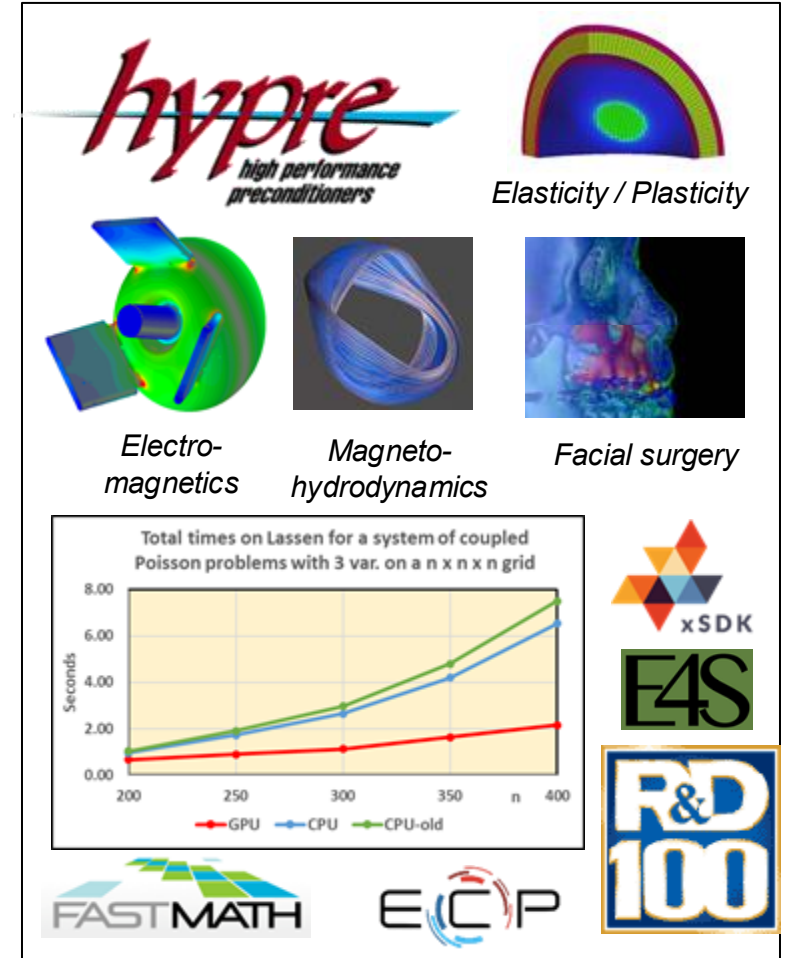
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Highly scalable multilevel solvers and preconditioners. Unique user-friendly interfaces. Flexible software design. Used in a variety of applications. Freely available.

- **Conceptual interfaces**
 - Structured, semi-structured, finite elements, linear algebraic interfaces
 - Provide natural “views” of the linear system
 - Provide for efficient (scalable) linear solvers through effective data storage schemes
- **Scalable preconditioners and solvers**
 - Structured and unstructured algebraic multigrid solvers
 - Maxwell solvers, H-div solvers
 - Multigrid solvers for nonsymmetric systems: pAIR
 - Multigrid reduction (MGR) for systems of PDEs
 - Matrix-free Krylov solvers
 - ILU and FSAI preconditioners
- **Exascale early systems GPU-readiness**
 - Nvidia GPU (CUDA), AMD GPU (HIP), Intel GPU (SYCL)
- **Open-source software**
 - Used worldwide in a vast range of applications
 - Can be used through PETSc and Trilinos
 - Available on github: <https://www.github.com/hypre-space/hypre>



<http://www.llnl.gov/CASC/hypre>

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SuperLU



Supernodal Sparse LU Direct Solver. Flexible, user-friendly interfaces.
Examples show various use scenarios. Testing code for unit-test. BSD license.

Capabilities

- Serial (thread-safe), shared-memory (SuperLU_MT, OpenMP or Pthreads), distributed-memory (SuperLU_DIST, hybrid MPI+ OpenM + CUDA/HIP).
 - Written in C, with Fortran interface
- Sparse LU decomposition (can be nonsymmetric sparsity pattern), triangular solution with multiple right-hand sides
- Incomplete LU (ILUTP) preconditioner in serial SuperLU
- Sparsity-preserving ordering: minimum degree or graph partitioning applied to $A^T A$ or $A^T + A$
- User-controllable pivoting: partial pivoting, threshold pivoting, static pivoting
- Condition number estimation, iterative refinement, componentwise error bounds

Exascale early systems GPU-readiness

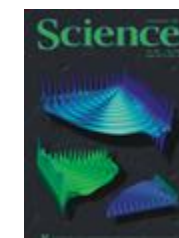
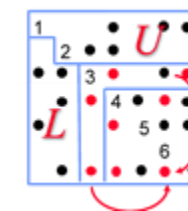
- Available: Nvidia GPU (CUDA), AMD GPU (HIP)
- In progress: Intel GPU (SYCL)

Parallel Scalability

- Factorization strong scales to 32,000 cores (IPDPS'18, JPDC'19)
- Triangular solve strong scales to 4000 cores (SIAM CSC'18, SIAM PP'20, SC'23)

Open-source software

- Used in a vast range of applications, can be used through PETSc and Trilinos, ...
- available on github



ITER tokamak quantum mechanics

Widely used in commercial software, including AMD (circuit simulation), Boeing (aircraft design), Chevron, ExxonMobile (geology), Cray's LibSci, FEMLAB, HP's MathLib, IMSL, NAG, SciPy, OptimaNumerics, Walt Disney Animation.



<https://portal.nersc.gov/project/sparse/superlu/>

STRUMPACK

Structured Matrix Package



Hierarchical solvers for dense rank-structured matrices and fast algebraic sparse solver and robust and scalable preconditioners.



■ Dense Matrix Solvers using Hierarchical Approximations

- Hierarchical partitioning, low-rank approximations
- Hierarchically Semi-Separable (HSS), Hierarchically Off-Diagonal Low-Rank (HODLR), Hierarchically Off-Diagonal Butterfly (HODBF), Block Low-Rank (BLR), Butterfly
- C++ Interface to ButterflyPACK (Fortran)
- Applications: BEM, Cauchy, Toeplitz, kernel & covariance matrices, ...
- Asymptotic complexity much lower than LAPACK/ScalAPACK routines

■ Sparse Direct Solver

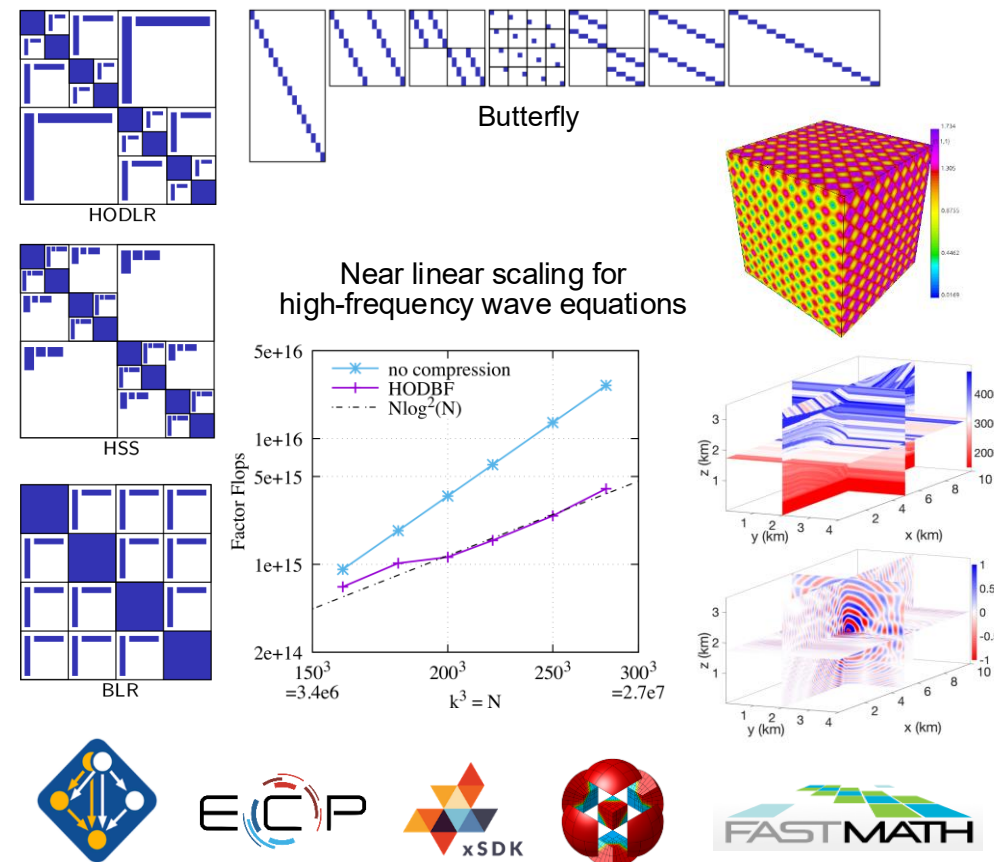
- Algebraic sparse direct solver
- GPU: CUDA, HIP/ROCm, DPC++ (in progress)
- Orderings: (Par)METIS, (PT)Scotch, RCM

■ Preconditioners

- Approximate sparse factorization, using hierarchical matrix approximations
- Scalable and robust, aimed at PDE discretizations, indefinite systems, ...
- Iterative solvers: GMRES, BiCGStab, iterative refinement

■ Software

- BSD license
- Interfaces from PETSc, MFEM, Trilinos, available in Spack



github.com/pghysels/STRUMPACK

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Adaptive time integrators for ODEs and DAEs and efficient nonlinear solvers
Used in a variety of applications. Freely available. Encapsulated solvers & parallelism.

■ ODE and DAE time integrators:

- *CVODE*: adaptive order and step BDF (stiff) & Adams (non-stiff) methods for ODEs
- *ARKODE*: adaptive step implicit, explicit, IMEX, and multirate Runge-Kutta methods for ODEs
- *IDA*: adaptive order and step BDF methods for DAEs
- *CVODES* and *IDAS*: provide forward and adjoint sensitivity analysis capabilities

■ Nonlinear Solvers: *KINSOL* – Newton-Krylov; accelerated Picard and fixed point

■ Modular Design: Easily incorporated into existing codes; Users can supply their own data structures and solvers or use SUNDIALS provided modules

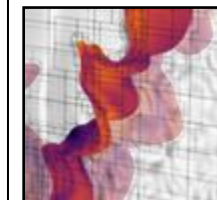
■ Support on NVIDIA, AMD, and Intel GPUs:

- Vectors: CUDA, HIP, OpenMP Offload, RAJA, SYCL (DPC++)
- Linear solvers: cuSOLVER, MAGMA, matrix-free Krylov methods

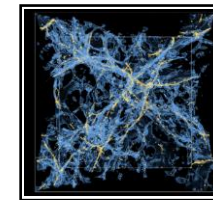
■ Open Source: BSD License; Download from LLNL site, GitHub, or Spack

- Supported by extensive documentation; user email list with an active community
- Available through MFEM, AMReX, deal.II, and PETSc

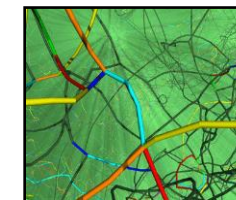
SUNDIALS is used worldwide in applications throughout research and industry



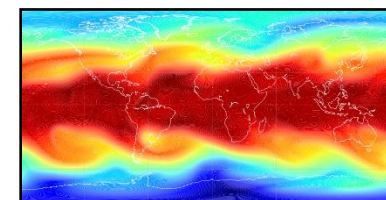
*Combustion
(Pele)*



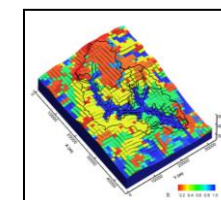
*Cosmology
(Nyx)*



*Dislocation dynamics
(ParaDiS)*



*Atmospheric Dynamics
(Tempest)*

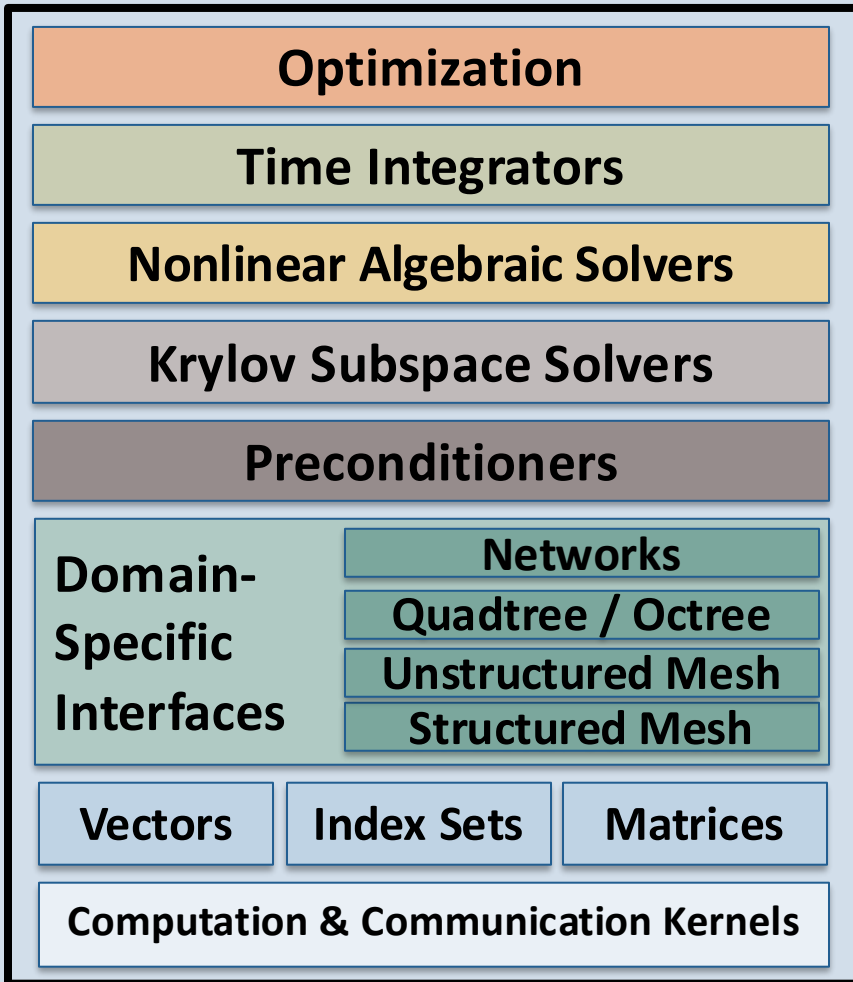


*Subsurface flow
(ParFlow)*



<http://www.llnl.gov/casc/sundials>

Scalable algebraic solvers for PDEs. Encapsulate parallelism in high-level objects. Active & supported user community. Full API from Fortran, C/C++, Python.

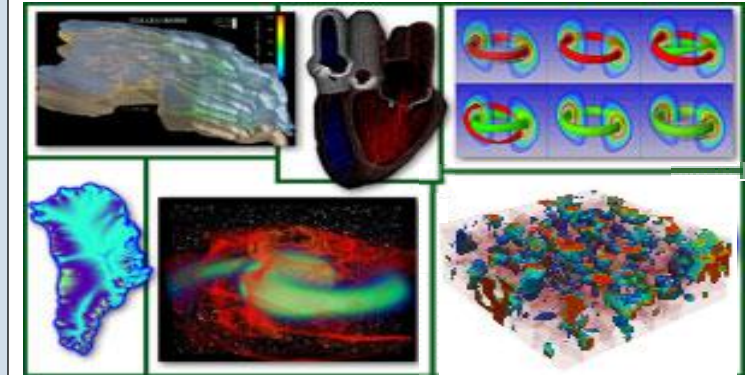


■ **Easy customization and composability of solvers at runtime**

- Enables optimality via flexible combinations of physics, algorithmics, architectures
- Try new algorithms by composing new/existing algorithms (multilevel, domain decomposition, splitting, etc.)

■ **Portability & performance**

- Largest DOE machines, also clusters, laptops; NVIDIA, AMD, and Intel GPUs
- Thousands of users worldwide



PETSc provides the backbone of diverse scientific applications.
clockwise from upper left: hydrology, cardiology, fusion, multiphase steel, relativistic matter, ice sheet modeling



<https://www.mcs.anl.gov/petsc>

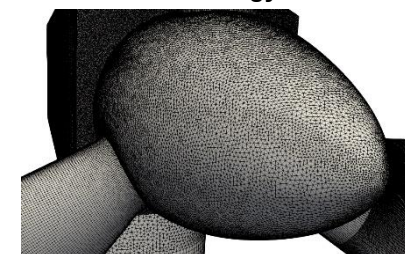
Agenda

<https://extremecomputingtraining.anl.gov/agenda-2025/#Track-5>

Time	Room?	Room?
8:30 – 9:30	Introduction to Numerical Software – Ulrike Yang	
9:30 – 10:45	Structured Discretization (AMReX) – Andrew Myers, Weiqun Zhang	Unstructured Discretization (MFEM/PUMI) – Mark Shephard, Cameron Smith, Mark Stowell
10:45 – 11:15	Break, Subject Matter Expert (SME) Selections, Panel Questions	
11:15 – 12:30	Iterative Solvers & Algebraic Multigrid (hypre) – Daniel Osei-Kuffuor, Ulrike Yang	Direct Solvers (SuperLU, STRUMPACK) – Sherry Li, Yang Liu
12:30 – 1:30	Lunch, SME Selections, Panel Questions	
1:30 – 2:45	Nonlinear Solvers (PETSc) – Richard Mills	Time Integration (SUNDIALS) – David Gardner
2:45 – 3:15	Break, SME Selections, Panel Questions Due	
3:15 – 4:30	Optimization (TAO) – Toby Isaac	Iterative Solvers & Algebraic Multigrid (Trilinos/ Belos/MueLU) – Christian Glusa, Graham Harper
4:30 – 5:15	Panel: Extreme-Scale Numerical Algorithms and Software	
5:15 – 6:30	Optional Activity: SME Speed-dating	
6:30 – 7:30	Dinner	
7:30 – 9:00	After-Dinner Talk: Jim Demmel	

Iterative Krylov-based solvers. Templated C++ allows for generic scalar, ordinal, and compute node types.

- Sandia National Laboratories is a multimission laboratory managed and operated by National Technology & Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International Inc. for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.



Thomas et al., "High-fidelity simulation of wind- turbine incompressible flows", SISC, 2019.



<https://trilinos.github.io/belos.html>

Trilinos/MueLu

Structured and unstructured aggregation-based algebraic multigrid (AMG) preconditioners

- **Robust, scalable, portable AMG preconditioning critical for many large-scale simulations**

- Multifluid plasma simulations
- Shock physics
- Magneto-hydrodynamics (MHD)
- Low Mach computational fluid dynamics (CFD)



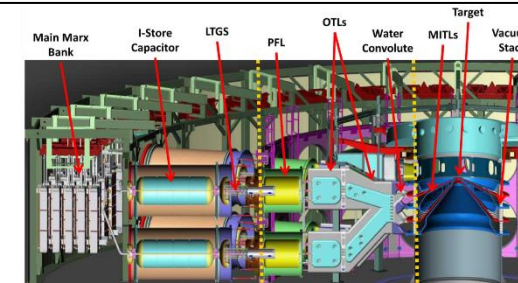
- **Capabilities**

- Aggregation-based coarsening
- **Smoothers**: Jacobi, GS, /1 GS, polynomial, ILU, sparse direct
- **Load-balancing** for good parallel performance
- Structured coarsening, geometric multigrid
- Setup and solve phases can run on GPUs.
- Performance portability via Kokkos (CPUs, NVIDIA/Intel/AMD GPUs, Xeon Phi)

- **Research Areas**

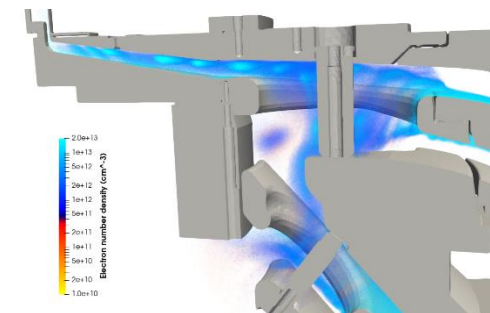
- AMG for multiphysics
- Multigrid for coupled structured/unstructured meshes
- Algorithm selection via machine learning

Sandia National Laboratories is a multimission laboratory managed and operated by National Technology & Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.



Z machine diagram, from "Redesign of a High Voltage Test Bed for Marxes on Z", W.M. White et al., 2018.

AMG preconditioning for H(curl) systems is key enabling technology in Z machine simulations for determining power from Marx banks to Target.

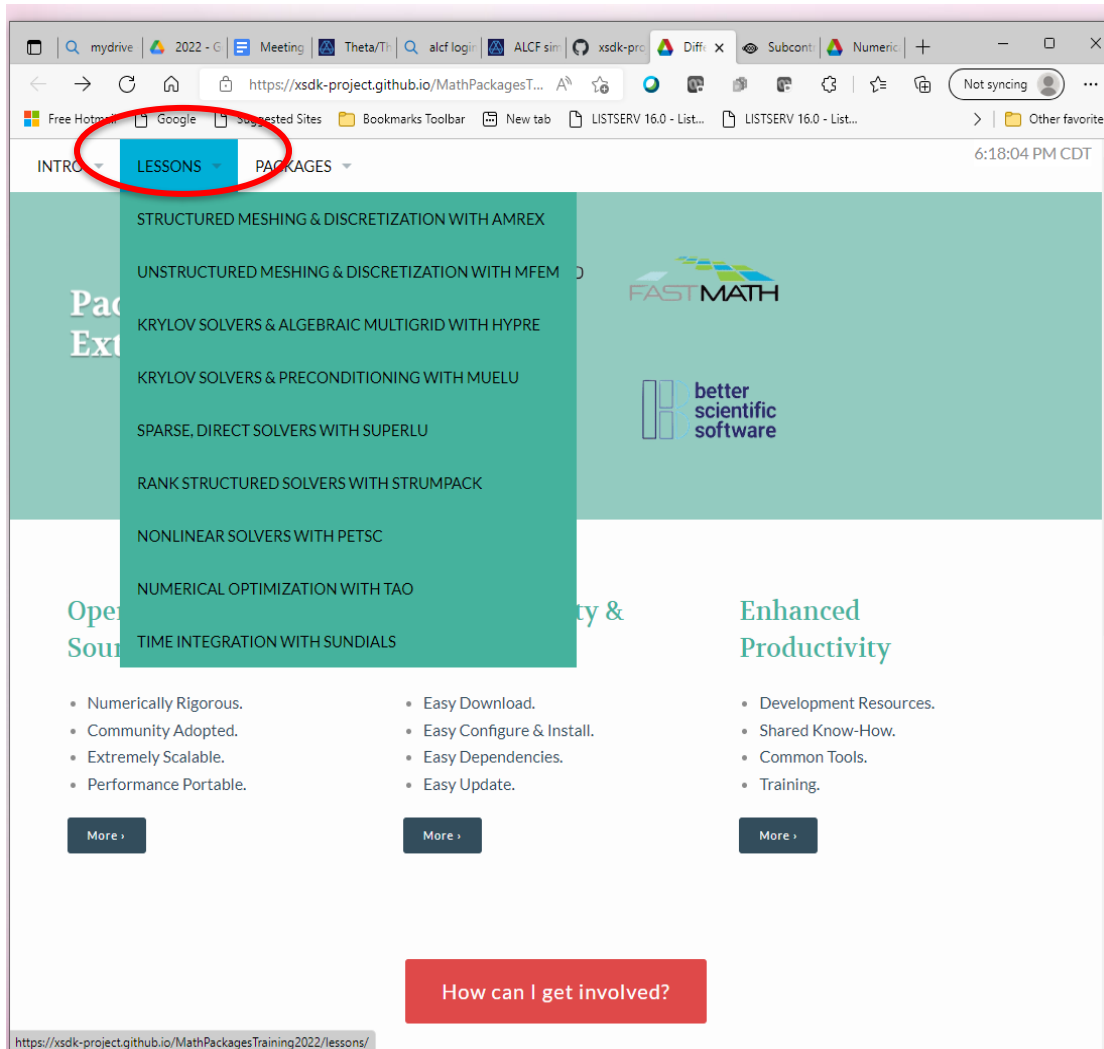


Plasma density in Z machine Target simulation, courtesy of D. Sirajuddin (SNL).



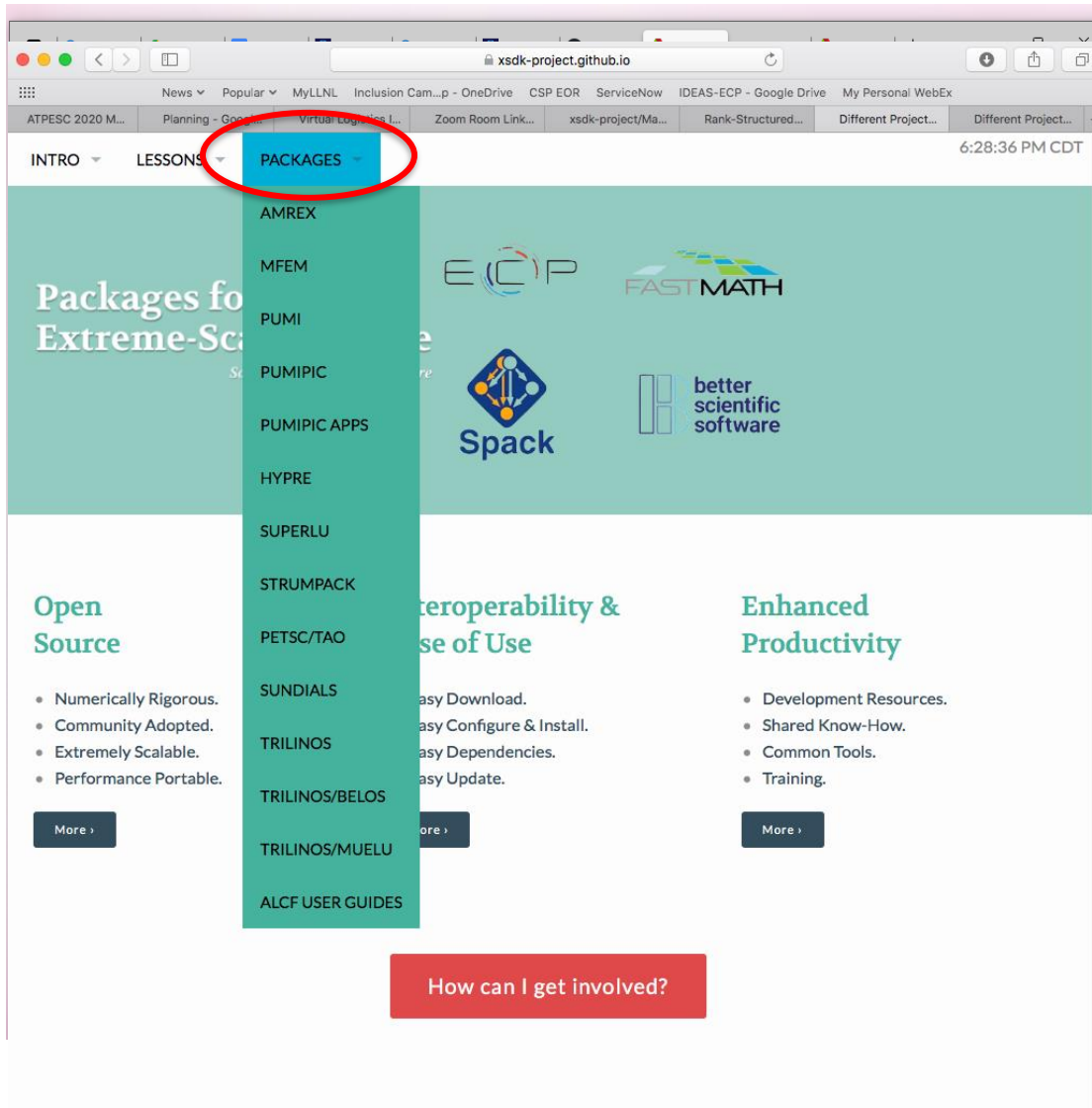
<https://trilinos.github.io/muelu.html>

<https://xsdk-project.github.io/MathPackagesTraining2025/>



- Hands-on Lessons

<https://xsdk-project.github.io/MathPackagesTraining2025/>



- Hands-on Lessons
- Packages

Agenda

<https://extremecomputingtraining.anl.gov/agenda-2025/#Track-5>

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7:30 – 9:00	After-Dinner Talk: Jim Demmel	

Panel: Extreme-Scale Numerical Algorithms and Software

- **Q&A Session:** ATPESC learners ask questions about working with numerical packages and the community of numerical package developers
 - Questions in **#numerical** slack channel and via [Question Submission Form](#)

- Panelists



Andrew Myers, LBL



Ulrike Yang, LLNL



Mark Stowell, LLNL



Yang Liu, LBL

- Moderator



Toby Isaac, Nvidia

Panel Question Submission Form

Please enter here a question you would like to ask our panelists during the 45 minute panel session.

We ask that you please include your name in case we may need to call upon you to clarify your question.

Next steps: <https://xsdk-project.github.io/MathPackagesTraining2025/agenda>

- **WrapUp (Toby Isaac) @ 4:30pm**

Panel: Main Room @ 4:45 pm

- **SME Speed Dating: @7:30pm**

- **During breaks and lunch**

- **Provide Panel Questions**

Due: 4:30 pm

- **Sign up for discussions with numerical software developers (optional)**

- **Your email address**

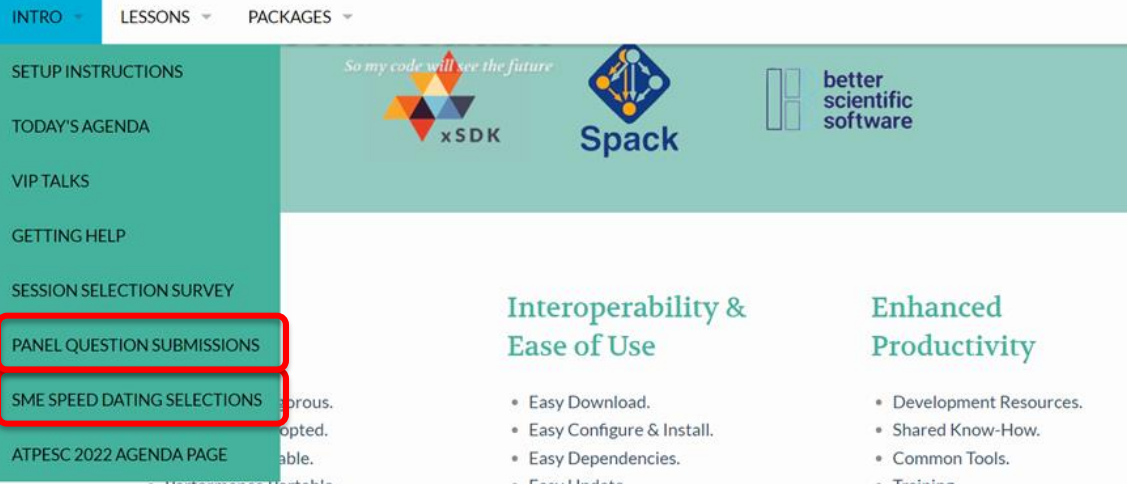
Due 5:00 pm



Subject Matter Expert (SME) 2-on-1 interviews

This is an optional activity. It is a great opportunity to spend some time chatting with various subject matter experts (SMEs).

In the form below, you may enter your first, second and third priorities for up to three, 20 minute, two-on-one discussions with various SMEs during the evening session.



The screenshot shows the xSDK website. The navigation menu on the left includes: INTRO, LESSONS, PACKAGES, SETUP INSTRUCTIONS, TODAY'S AGENDA, VIP TALKS, GETTING HELP, SESSION SELECTION SURVEY, **PANEL QUESTION SUBMISSIONS**, **SME SPEED DATING SELECTIONS**, and ATPESC 2022 AGENDA PAGE. The header features the slogan "So my code will see the future" and logos for xSDK, Spack, and better scientific software. The main content area highlights "Interoperability & Ease of Use" and "Enhanced Productivity" with lists of benefits.

Interoperability & Ease of Use

- Easy Download.
- Easy Configure & Install.
- Easy Dependencies.
- Easy Update.

Enhanced Productivity

- Development Resources.
- Shared Know-How.
- Common Tools.
- Training.

- Tip: Consider setting Preferences to customize when to send

The screenshot shows the 'Preferences' window with the 'Advanced' tab selected. Under the 'Input options' section, the option 'When writing a message, press' is circled in red. This option has two radio buttons: 'Send the message' (unselected) and 'Start a new line (use `⌘` Enter to send)' (selected). Other options visible include 'When typing code with ```', 'Format messages with markup', and 'Messages & media'.



VIPs of ATPESC Extreme-Scale Numerical Software Track



- **Paul Fisher, Argonne National Laboratory** [[bio](#)]
 - **Scaling Computational Fluid Dynamics to Exascale and Beyond**
 - ATPESC 2025, Monday, July 28, 7:30pm



- **Jack Dongarra, Univ of Tennessee** [[bio](#)]
 - **Growing up at Argonne National Laboratory**
 - ATPESC 2025, Monday, August 4, 7:30pm
 - Adaptive Linear Solvers and Eigensolvers, ATPESC 2019 [[video](#)]



- **Jim Demmel, UC Berkeley** [[bio](#)]
 - **After Dinner Talk**
 - ATPESC 2025, Wednesday, August 5, 7:30pm
 - ENLA Seminar, June 2020 [[video](#)]



The ATPESC Team 2024

Extreme-scale numerical algorithms and software
Integrated lectures and hands-on examples, panel session, individual discussions ... and more!



Toby Isaac, Nvidia



Andrew Myers, LBL



Richard Mills, ANL



Mark Shephard, RPI



Cameron Smith, RPI



Ulrike Yang, LLNL



Christian Glusa, SNL



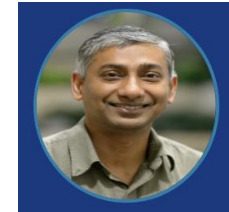
Graham Harper, SNL



Sherry Li, LBL



Mark Stowell, LLNL



Satish Balay, ANL



Daniel Osei-Kuffuor, LLNL



David Gardner, LLNL



Weiqun Zhang, LBL



Yang Liu, LBL

Track 5: Numerical Algorithms and Software: Tutorial Goals

1.

Provide a basic understanding of a variety of applied mathematics algorithms for scalable linear, nonlinear, and ODE solvers, as well as discretization technologies (e.g., adaptive mesh refinement for structured and unstructured grids) and numerical optimization

2.

Provide an overview of software tools available to perform these tasks on HPC architectures ... including where to go for more info

3.

Practice using one or more of these software tools on basic demonstration problems

This presentation provides a high-level introduction to HPC numerical software

- How HPC numerical software addresses challenges in computational science and engineering (CSE)
- Toward extreme-scale scientific software ecosystems
- Using and contributing: Where to go for more info

Why is this important for you?

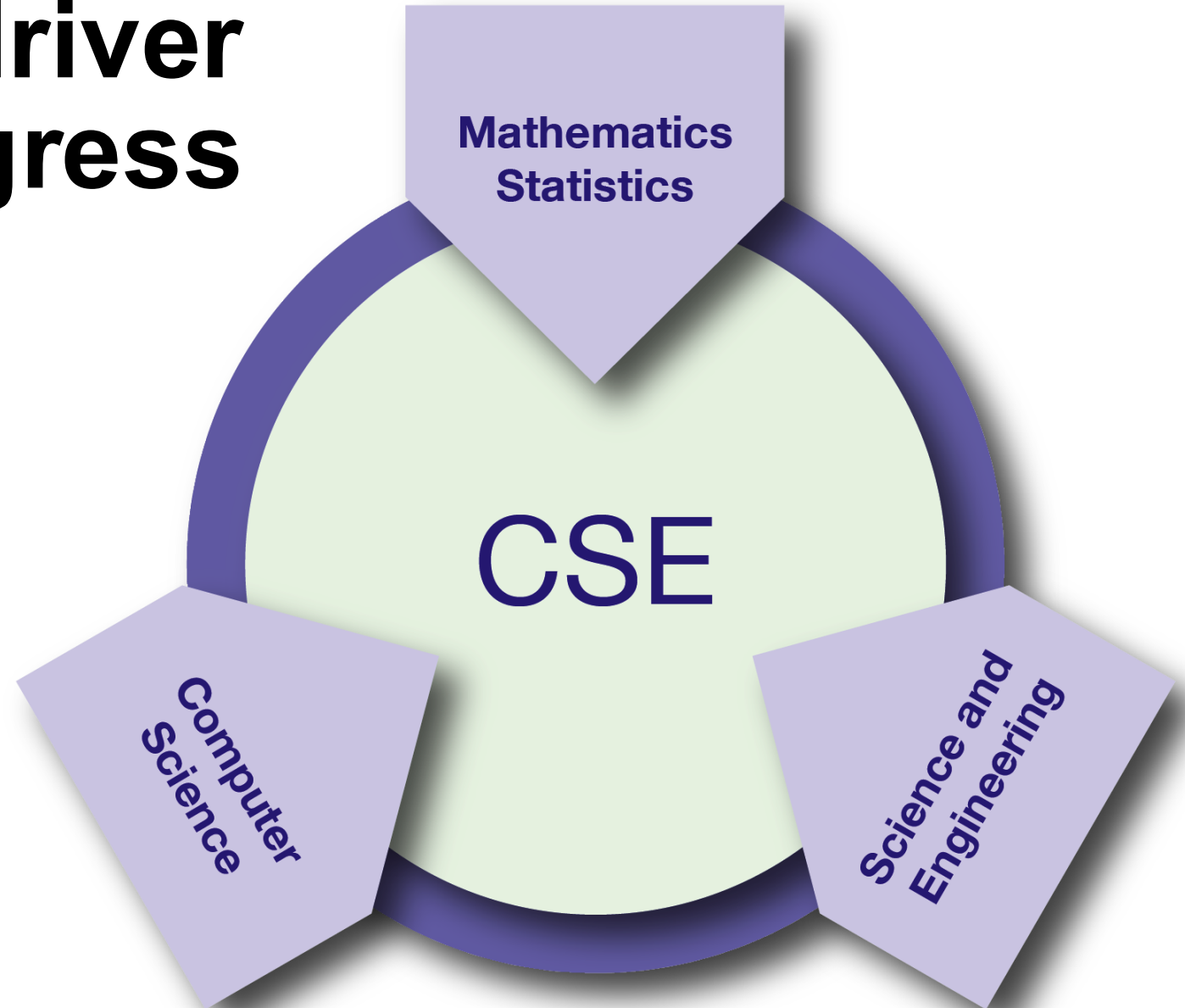
- Libraries enable users to focus on their primary interests
 - Reuse algorithms and data structures developed by experts
 - Customize and extend to exploit application-specific knowledge
 - Cope with complexity and changes over time
- More efficient, robust, reliable, scalable, sustainable scientific software
- Better science, broader impact of your work

CSE: Essential driver of scientific progress

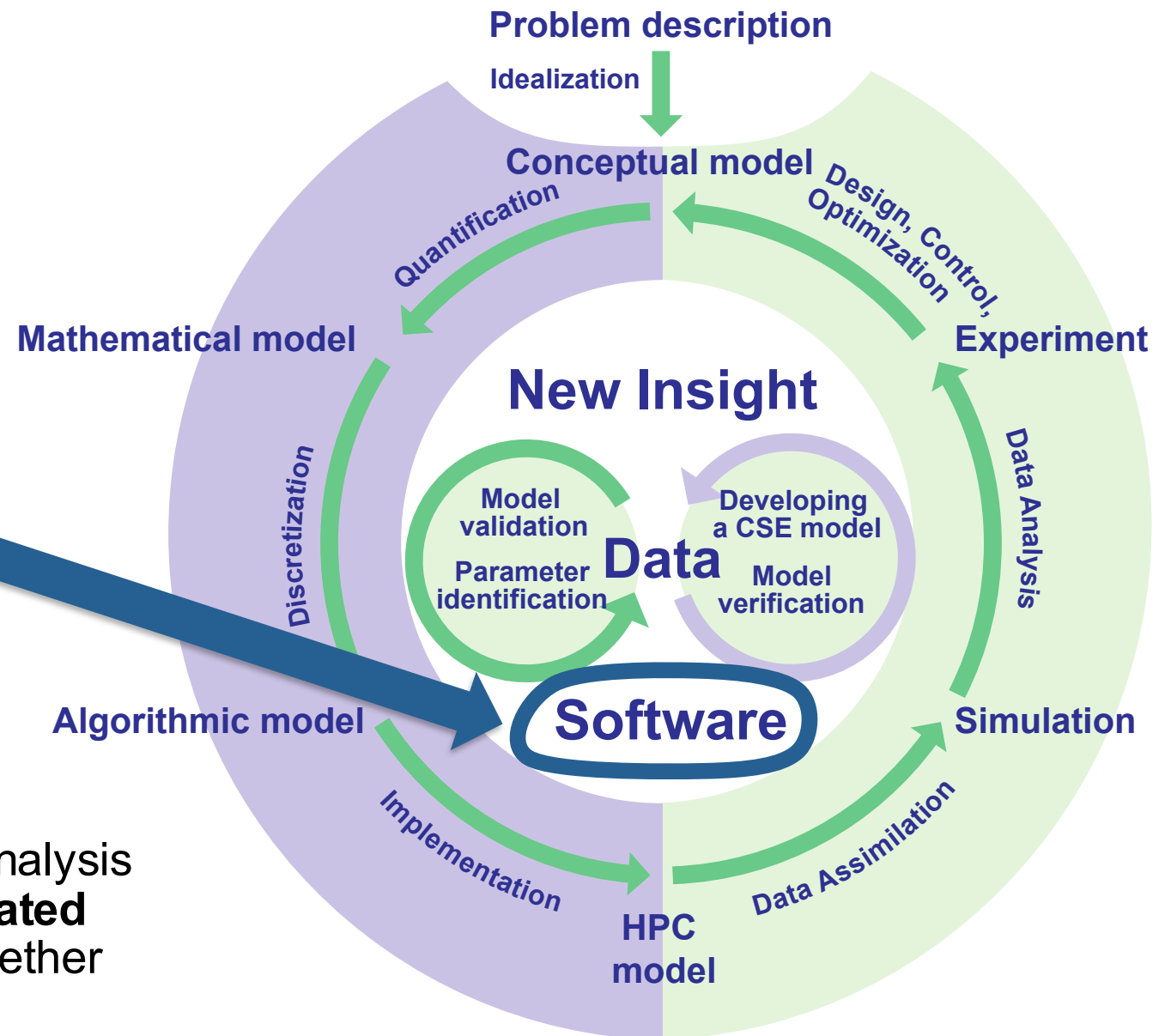
CSE = Computational Science & Engineering

Development and use of computational methods for scientific discovery

- all branches of the sciences
- engineering and technology
- support of decision-making across a spectrum of societally important applications



Software is the foundation of sustained CSE collaboration and scientific progress.



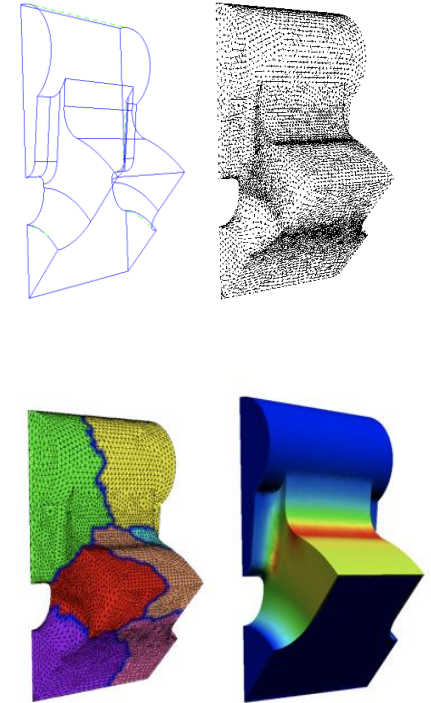
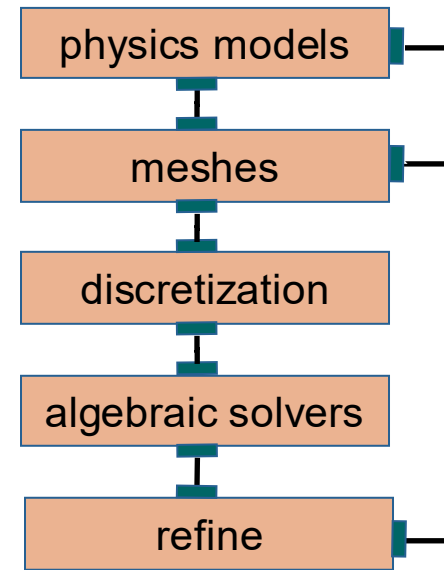
CSE cycle: Modeling, simulation, and analysis

- **Software: independent but interrelated elements** for various phases that together enable CSE

CSE simulation starts with a forward simulation that captures the physical phenomenon of interest

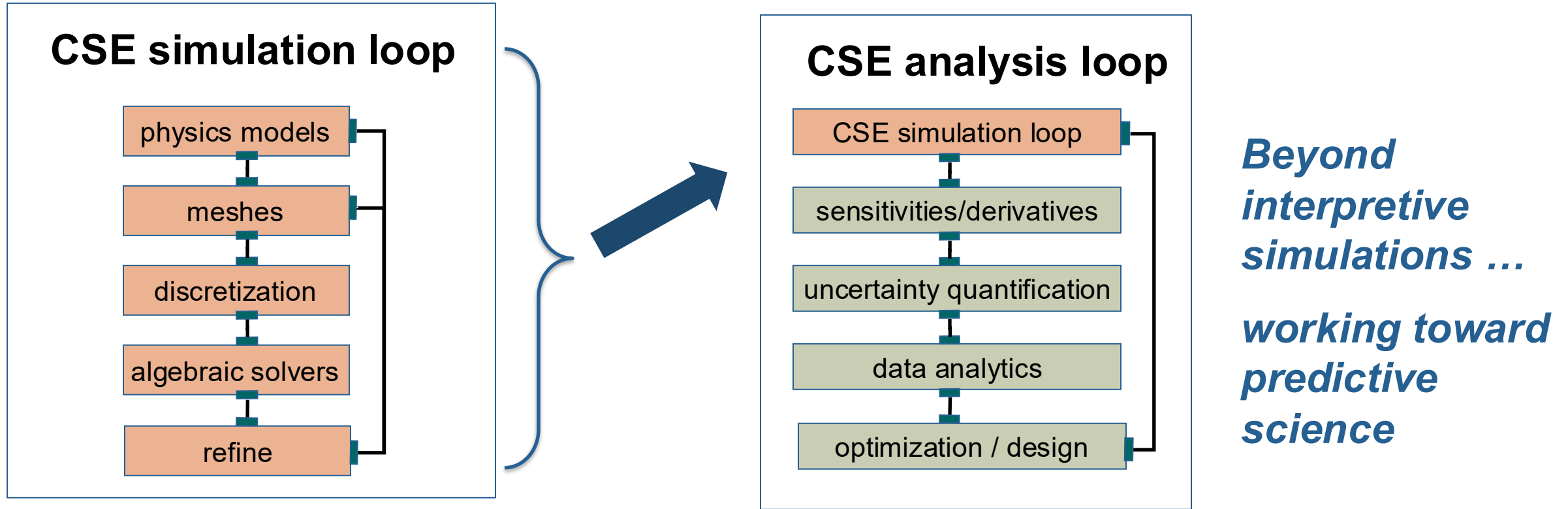
- Develop a mathematical model of the phenomenon of interest
- Approximate the model using a discrete representation
- Solve the discrete representation
- Adapt and refine the mesh or model
- Incorporate different physics, scales

CSE simulation loop



Requires: mesh generation, partitioning, load balancing, high-order discretization, time integration, linear & nonlinear solvers, eigensolvers, mesh refinement, multiscale/multiphysics coupling, etc.

CSE analysis builds on the CSE simulation loop ... and relies on even more numerical algorithms and software



Requires: adjoints, sensitivities, algorithmic differentiation, sampling, ensembles, data analytics, uncertainty quantification, optimization (derivative free & derivative based), inverse problems, etc.

First consider a very simple example

- 1D rod with one end in a hot water bath, the other in a cold water bath
- Mathematical model

$$\nabla^2 T = 0 \in \Omega$$
$$T(0) = 180^\circ \quad T(1) = 0^\circ$$

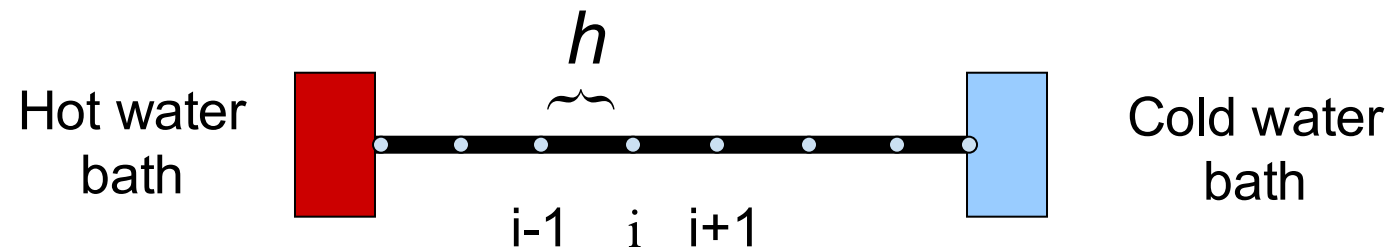


The first step is to discretize the equations

- Approximate the derivatives of the continuous equations with a discrete representation that is easier to solve
- One approach: Finite differences

$$\nabla^2 T \approx (T_{i+1} - 2T_i + T_{i-1})/h^2 = 0$$

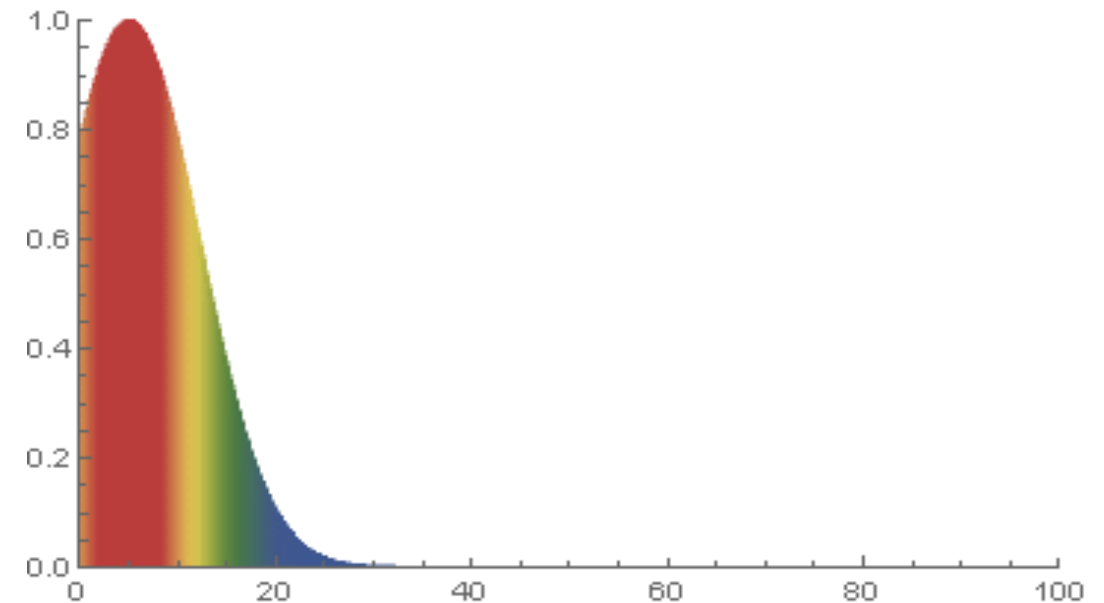
$$T_0 = 180^\circ \quad T_n = 0^\circ$$



Then you can solve for the unknowns T_i

- Set up a matrix of the unknown coefficients
 - include the known boundary conditions
- Solve the linear system for T_i

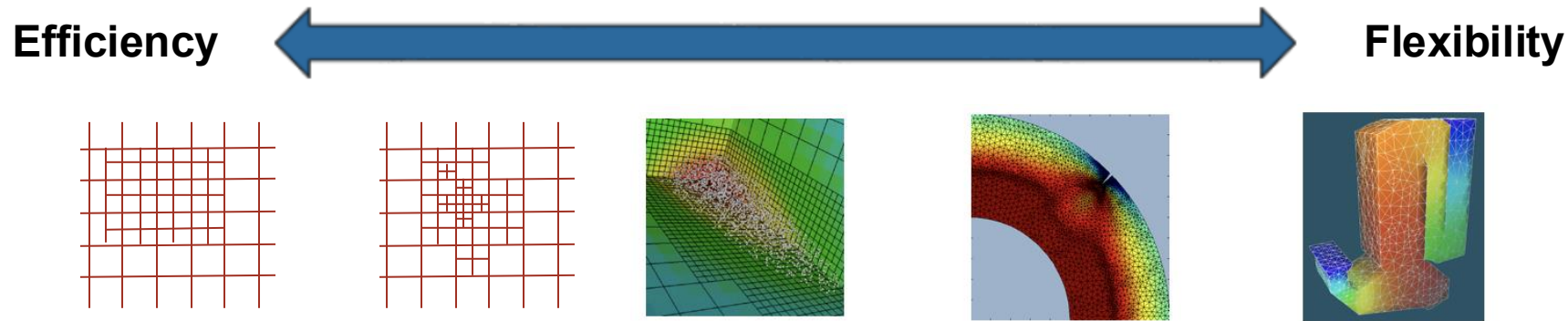
$$\begin{pmatrix} 2 & -1 & 0 & \dots\dots\dots 0 \\ -1 & 2 & -1 & 0 & \dots\dots\dots 0 \\ 0 & -1 & 2 & -1 & 0 & \dots\dots 0 \\ & & & \dots\dots\dots & & \\ 0 & \dots\dots\dots 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ \vdots \\ T_{n-1} \end{pmatrix} = \begin{pmatrix} 180 h^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$



- Visualize and analyze the results

As problems get more complicated, so do the steps in the process

- Different discretization strategies exist for differing needs



- Most problems are time dependent and nonlinear
 - Need higher algorithmic levels than linear solvers
- Increasingly combining multiple physical processes
 - Interactions require careful handling
- Goal-oriented problem solving requires optimization, uncertainty quantification

This work is founded on decades of experience and concerted team efforts to advance numerical software ...



<https://scidac5-fastmath.lbl.gov>



EXASCALE COMPUTING PROJECT

<https://exascaleproject.org>

- FASTMath SciDAC Institute
- Exascale Computing Project (ECP)
- Developers of xSDK packages

... While improving software productivity & sustainability as key aspects of advancing overall scientific productivity



<https://ideas-productivity.org>

- IDEAS Software Productivity Project
- Better Scientific Software Community

See also Track 8:
Software Sustainability (Aug 8)

Community efforts:
Join us!



<https://xsdk.info>



<https://bssw.io>



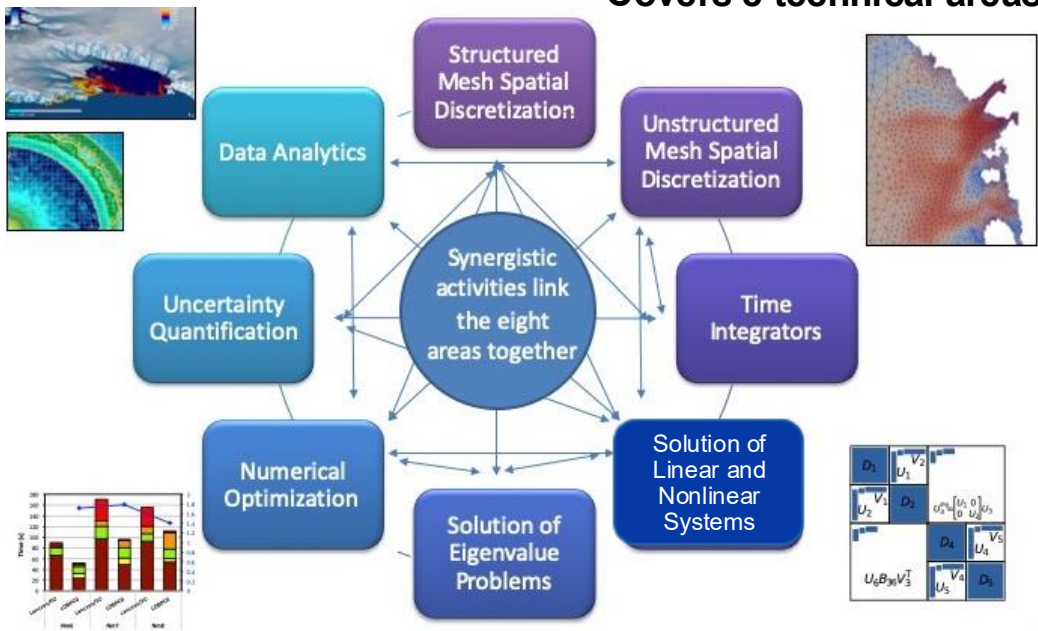
<https://e4s.io>

ATPESC 2025
EXTREME-SCALE COMPUTING

FASTMath: Frameworks, Algorithms & Scalable Technologies for Mathematics

<https://scidac5-fastmath.lbl.gov/>

Covers 8 technical areas



FASTMath Goals:

- Develop advanced numerical techniques for DOE applications
- Deploy high-performance software on DOE supercomputers
- Demonstrate basic research technologies from applied mathematics
- Engage and support of the computational science community

100's of person years of experience building math software

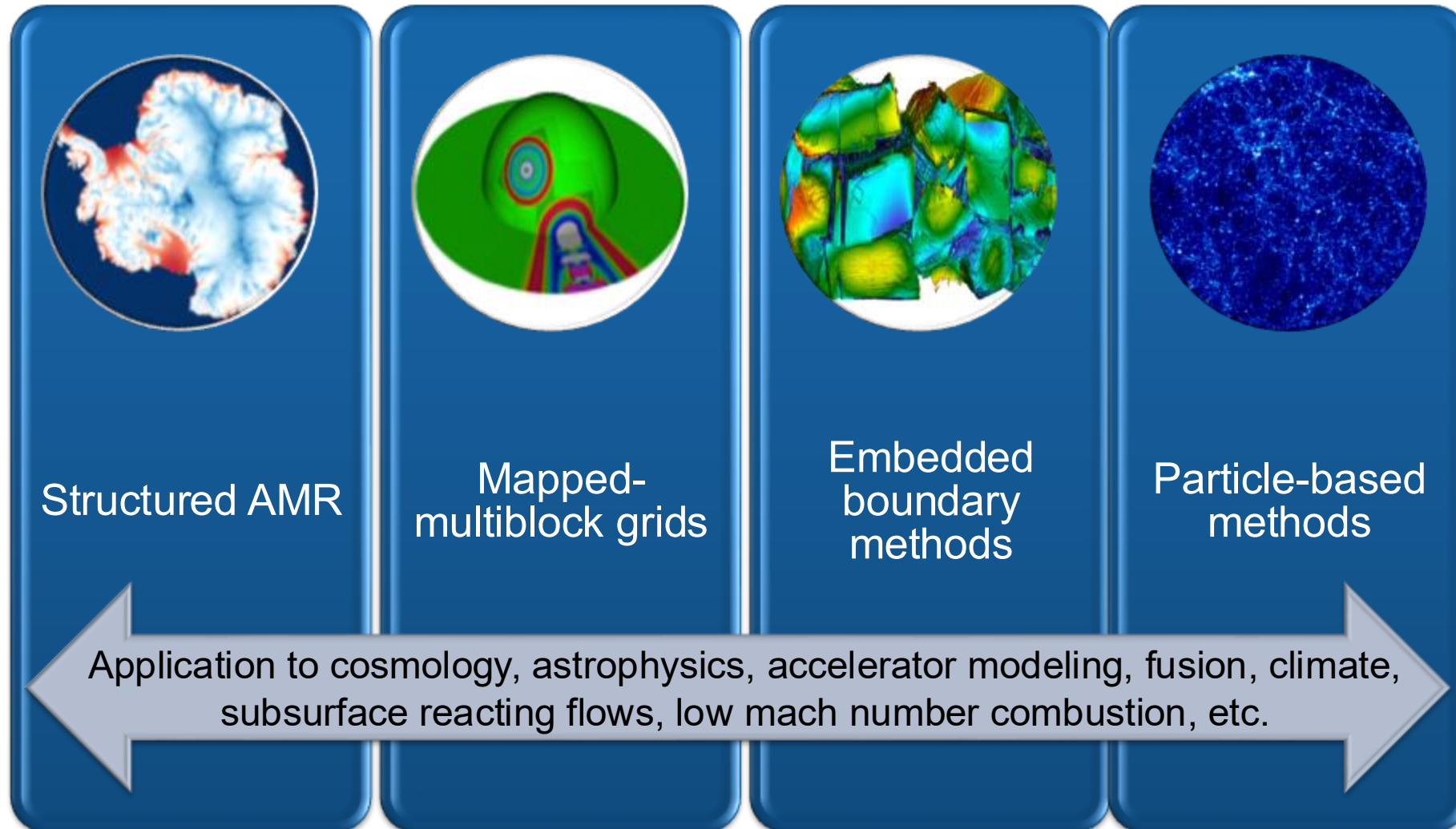
50+ researchers from 6 DOE labs and 6 universities



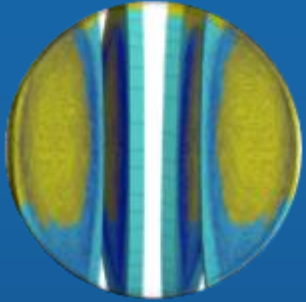
Carol Woodward, FASTMath Institute Director (cswoodward@llnl.gov)



Structured grid efforts focus on high-order, mapped grids, embedded boundaries, AMR, and particles



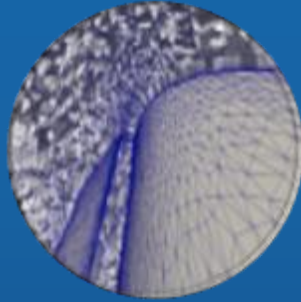
Unstructured grid capabilities focus on adaptivity, high-order, and the tools needed for extreme scaling



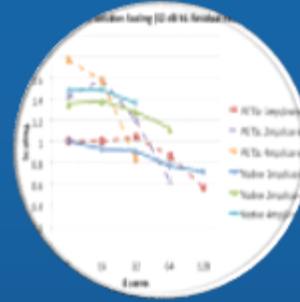
Parallel mesh infrastructures



Dynamic load balancing



Mesh adaptation and quality control



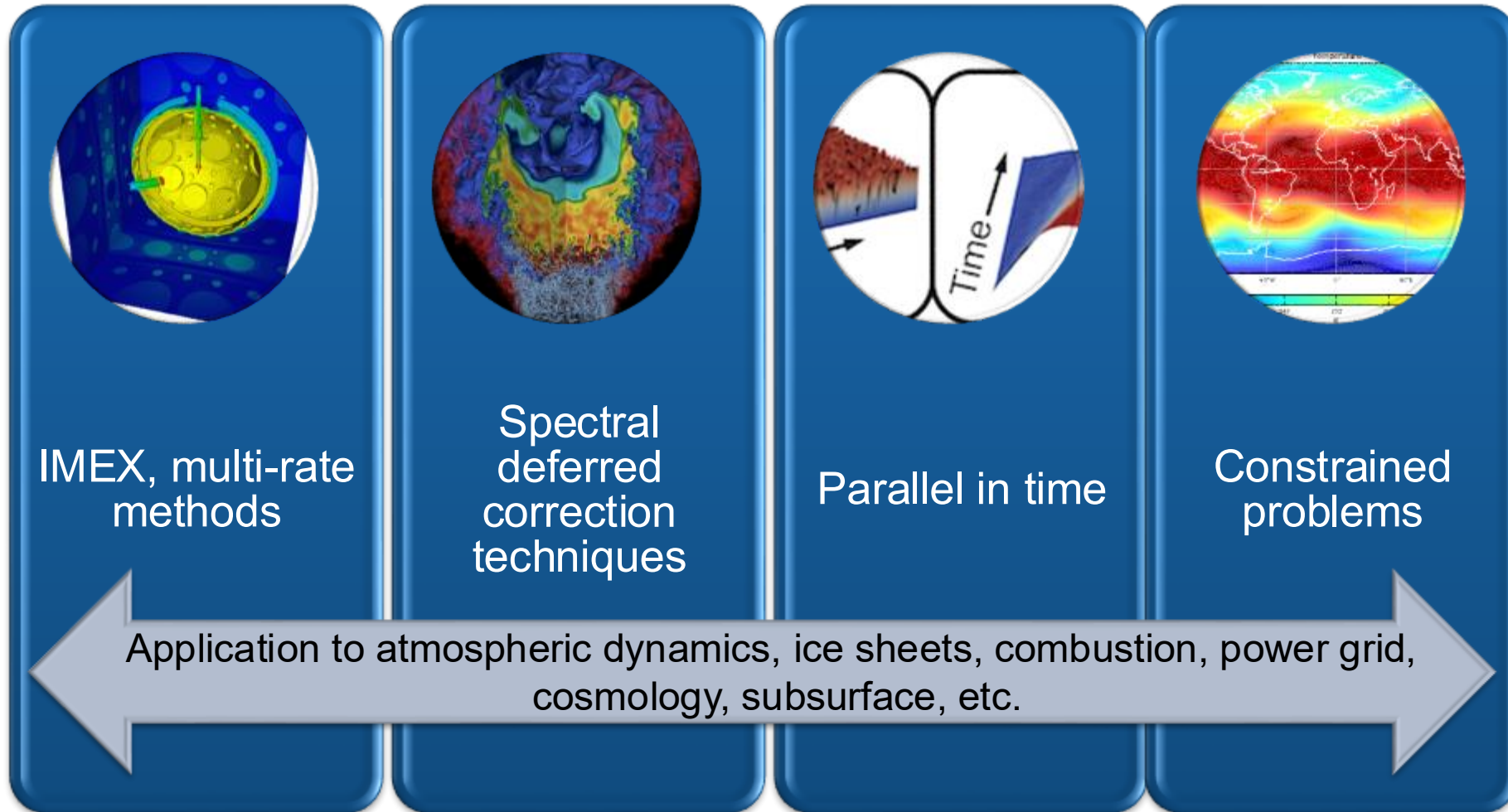
Parallel performance on unstructured meshes



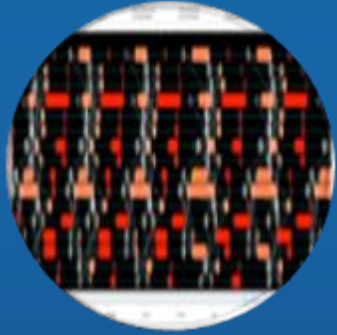
Architecture aware implementations

Application to fusion, climate, accelerator modeling, NNSA applications, nuclear energy, manufacturing processes, etc.

Time discretization methods provide efficient and robust techniques for stiff implicit, explicit and multi-rate systems



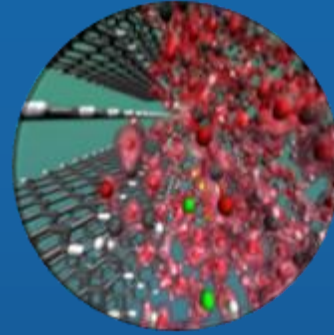
Research on algebraic systems provides key solution technologies to applications



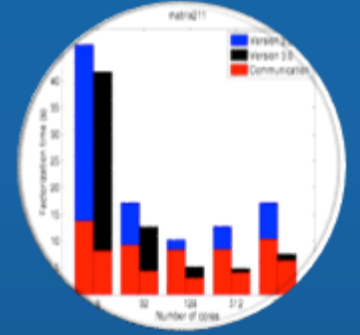
Linear system
solution using direct
and iterative solvers



Nonlinear system
solution using
acceleration
techniques and
globalized Newton
methods



Eigensolvers using
iterative techniques
and optimization



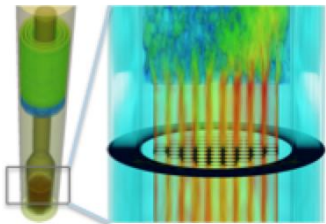
Architecture aware
implementations

Application to fusion, nuclear structure calculation, quantum chemistry,
accelerator modeling, climate, dislocation dynamics etc,

Multiphysics: A primary motivator for exascale

Multiphysics: greater than 1 component governed by its own principle(s) for evolution or equilibrium

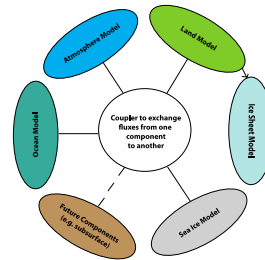
- Also: broad class of coarsely partitioned problems possess similarities



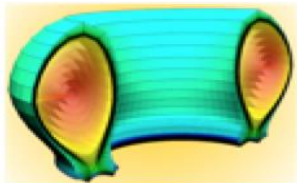
nuclear reactors
A. Siegel, ANL



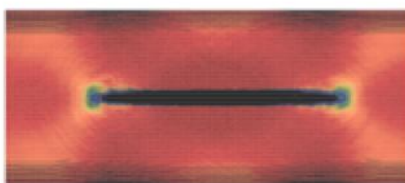
particle accelerators
K. Lee, SLAC



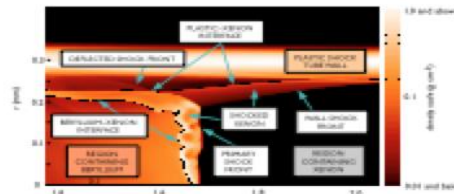
climate
K. Evans, ORNL



fusion
A. Hakim, PPPL



crack propagation
E. Kaxiras, Harvard



radiation hydrodynamics
E. Myra, Univ. of Michigan

IJHPCA, Feb 2013
Vol 27, Issue 1, pp. 4-83



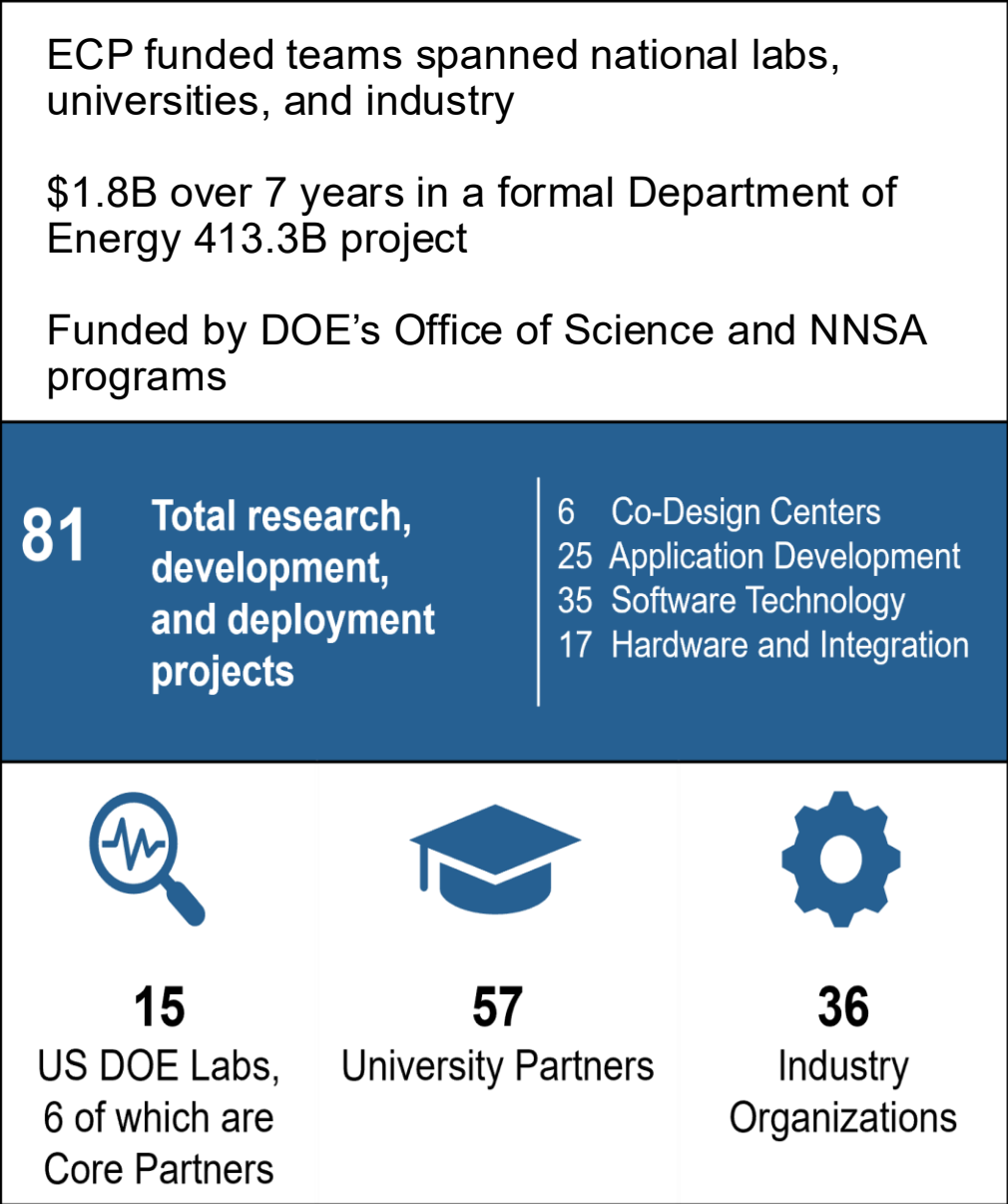
The International Journal of High Performance Computing Applications
27(1) 4-83
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DOI: 10.1177/1094342012468181
hpc.sagepub.com
SAGE

Multiphysics simulations: Challenges and opportunities

David E Keyes^{1,2}, Lois C McInnes³, Carol Woodward⁴, William Gropp⁵, Eric Myra⁶, Michael Pernice⁷, John Bell⁸, Jed Brown³, Alain Clo¹, Jeffrey Connors⁴, Emil Constantinescu³, Don Estep⁹, Kate Evans¹⁰, Charbel Farhat¹¹, Ammar Hakim¹², Glenn Hammond¹³, Glen Hansen¹⁴, Judith Hill¹⁰, Tobin Isaac¹⁵, Xiangmin Jiao¹⁶, Kirk Jordan¹⁷, Dinesh Kaushik³, Efthimios Kaxiras¹⁸, Alice Koniges⁸, Kihwan Lee¹⁹, Aaron Lott⁴, Qiming Lu²⁰, John Magerlein¹⁷, Reed Maxwell²¹, Michael McCourt²², Miriam Mehl²³, Roger Pawlowski¹⁴, Amanda P Randles¹⁸, Daniel Reynolds²⁴, Beatrice Rivière²⁵, Ulrich Rüde²⁶, Tim Scheibe¹³, John Shadid¹⁴, Brendan Sheehan⁹, Mark Shephard²⁷, Andrew Siegel³, Barry Smith³, Xianzhu Tang²⁸, Cian Wilson² and Barbara Wohlmuth²³

[doi:10.1177/1094342012468181](https://doi.org/10.1177/1094342012468181)

The Exascale Computing Project was designed to help launch the exascale era



- A **unique collaboration** brought together some of the brightest application, software, and computational experts from coast to coast
- **Best practices and lessons learned** for how to program GPUs – moving the nation forward
- **1000+ researchers** trained and ready for accelerator-based computing
- **1000+ students** introduced to HPC and Exascale computing through ECP's outreach, training, and workforce development initiatives

Technical work was largely complete as of Dec 31, 2023

Project leadership team is now working to close out the formal DOE 413.3B project

Next 18 slides courtesy of Lori Diachin

The Exascale Computing Project was started in 2016 and tasked to meet four DOE mission needs in high performance computing

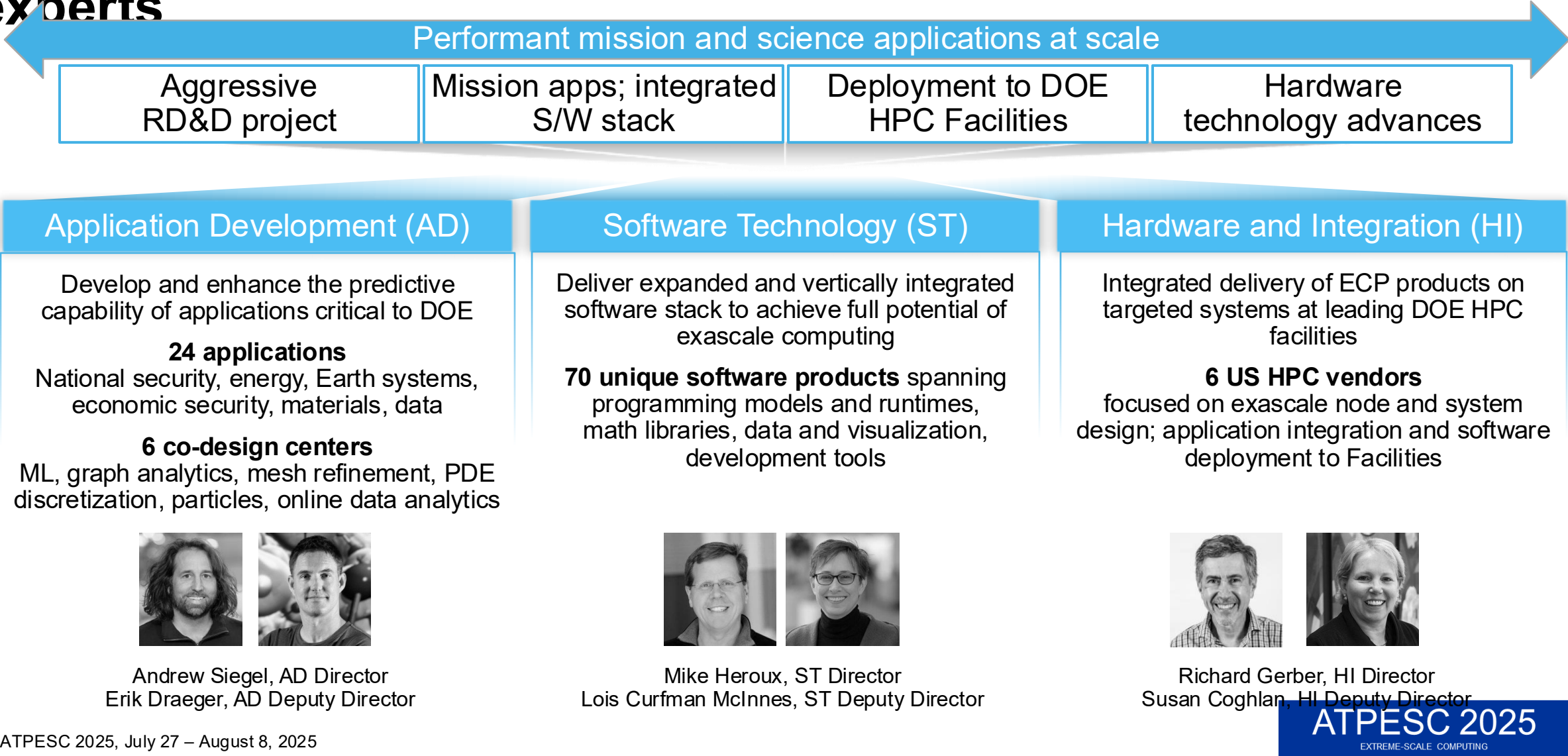
Deliver a long-term,
sustainable software ecosystem that can be
used and maintained for
years to come

Promote the **health of the
US HPC industry**

Ensure that exascale
systems can be used to
deliver **mission-critical
applications**

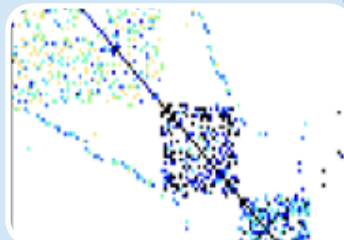
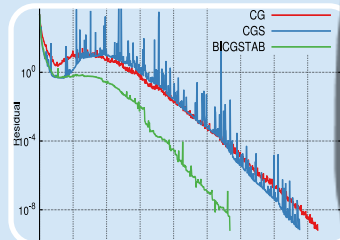
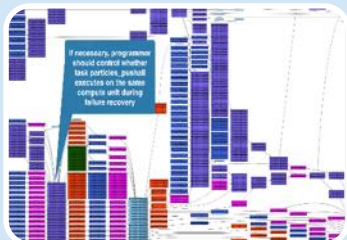
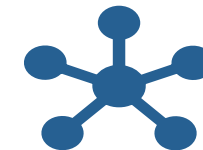
Maintain **international
leadership in HPC**

To meet mission needs, the ECP was organized into three technical focus areas partnered with formal project management experts



ECP invested in software technology areas

Emphasis for this presentation



Programming Models & Runtimes

- Enhance and get ready for exascale the MPI and OpenMP programming models (hybrid programming models, deep memory copies)
- Develop performance portability tools (e.g., Kokkos and Raja)
- Support alternate models for potential benefits and risk mitigation: PGAS (UPC++/GASNet), task-based models (Legion, PaRSEC)
- Libraries for deep memory hierarchy and power management



Rajeev Thakur

Development Tools

- Continued, multifaceted capabilities in portable, open-source LLVM compiler ecosystem to support expected ECP architectures, including support for F18
- Performance analysis tools that accommodate new architectures, programming models, e.g., PAPI, Tau



Jeff Vetter

Math Libraries

- Linear algebra, iterative linear solvers, direct linear solvers, integrators and nonlinear solvers, optimization, FFTs, etc
- Performance on new node architectures; extreme strong scalability
- Advanced algorithms for multi-physics, multiscale simulation and outer-loop analysis
- Increasing quality, interoperability, complementarity of math libraries



Sherry Li

Data and Visualization

- I/O via the HDF5 API
- Insightful, memory-efficient in-situ visualization and analysis
- Data reduction via scientific data compression
- Checkpoint restart



Jim Ahrens

Software Ecosystem

- Develop features in Spack necessary to support ST products in E4S, and the AD projects that adopt it
- Develop Spack stacks for reproducible turnkey software deployment
- Optimization and interoperability of containers for HPC
- Regular E4S releases of the ST software stack and SDKs with regular integration of new ST products



Todd Munson

NNSA ST

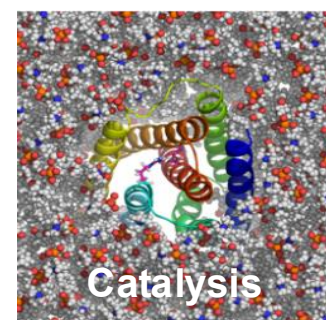
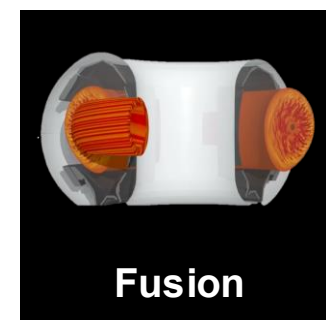
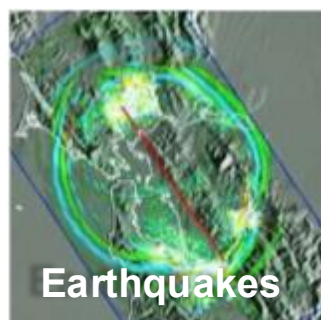
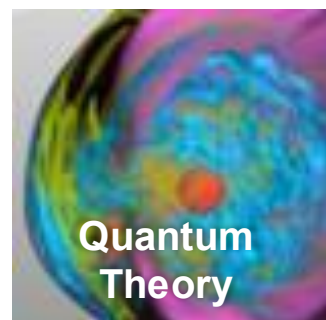
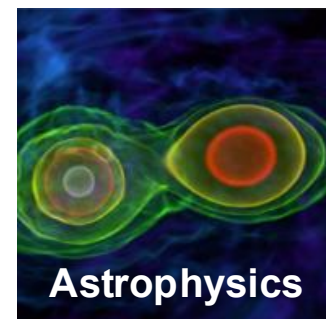
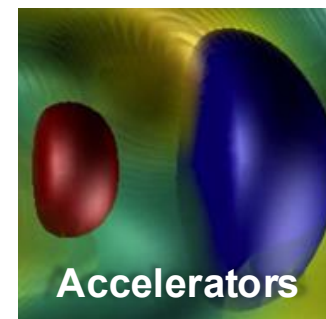
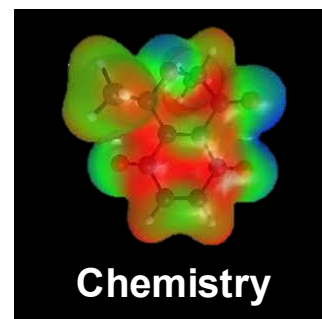
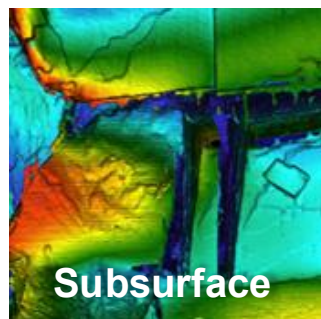
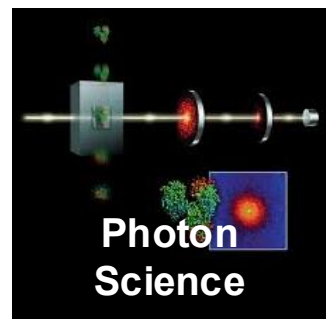
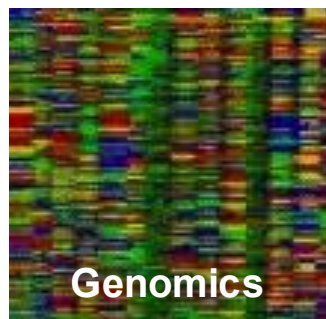
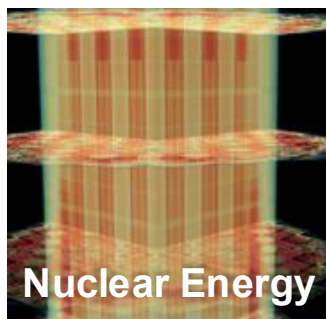
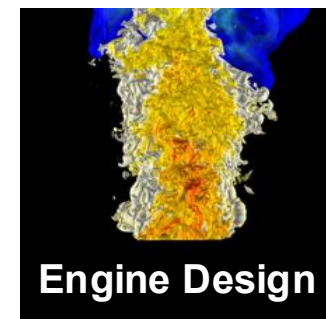
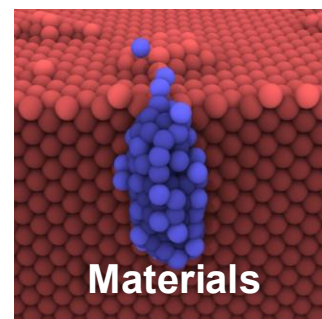
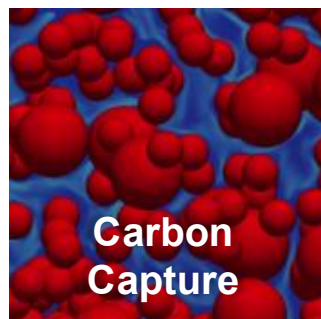
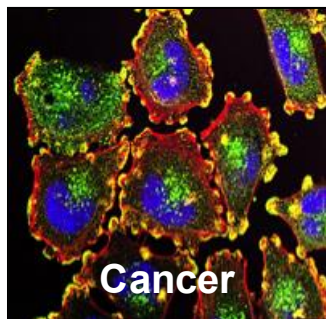
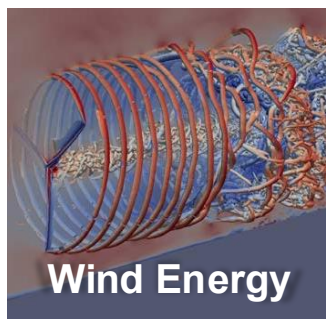
- Open source NNSA Software projects
- Projects that have both mission role and open science role
- Major technical areas: New programming abstractions, math libraries, data and viz libraries
- Cover most ST technology areas
- Subject to the same planning, reporting and review processes



Kathryn Mohror

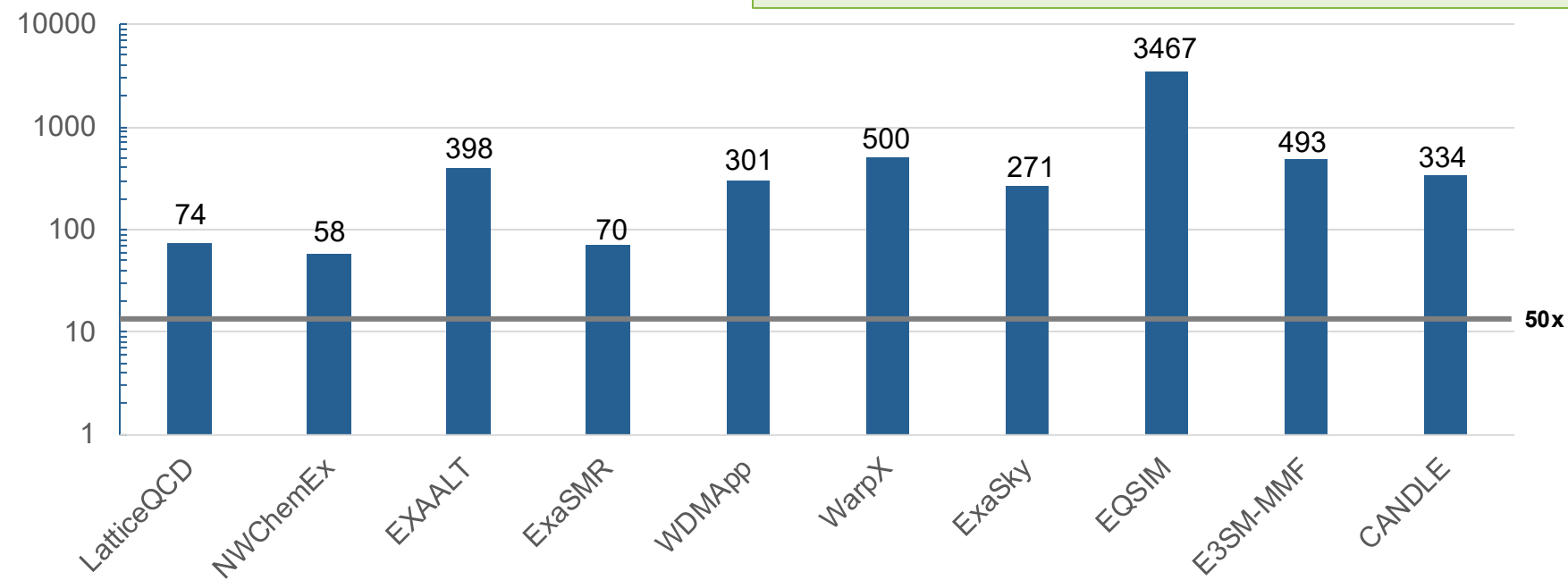
Area Leads:

ECP invested in a broad range of critical application areas; some with limited HPC experience at project inception



ECP application results exceeded expectations

10 out of 11 projects surpassed an ambitious 50x performance target



Co-design played a critical role

AMReX

CEED

CoPA

CODAR

ExaGraph

ExaLearn

9 out of 10 new HPC science projects completed exascale capability demonstrations

3 out of 4 NNSA applications demonstrated exascale readiness

GAMESS

ExaAM

ExaWind

ExaSGD

Subsurface

Combustion

MFIX-Exa

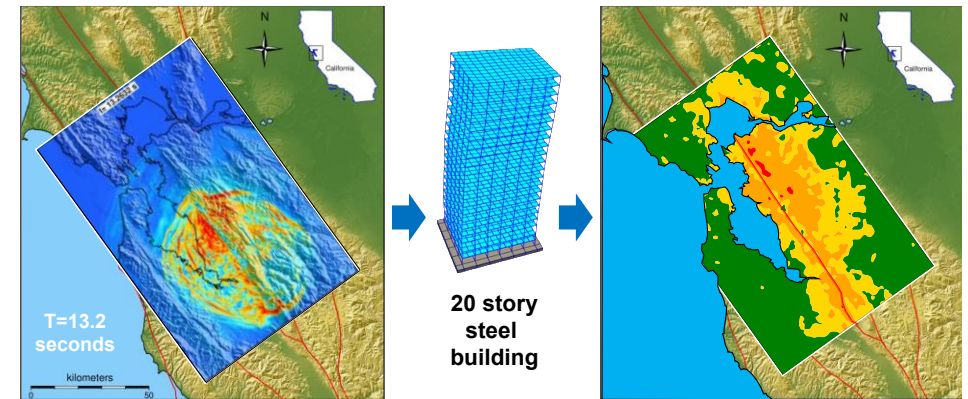
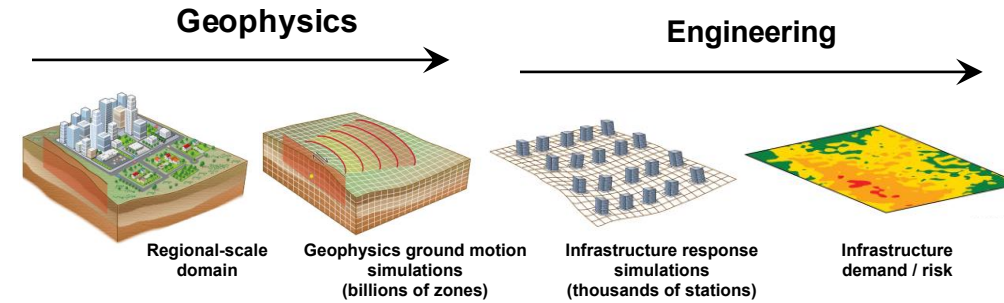
ExaStar

ExaBiome

One example EQSIM: exascale-capable code redesign had massive impact

EQSIM PI: Dave McCallen, LBNL

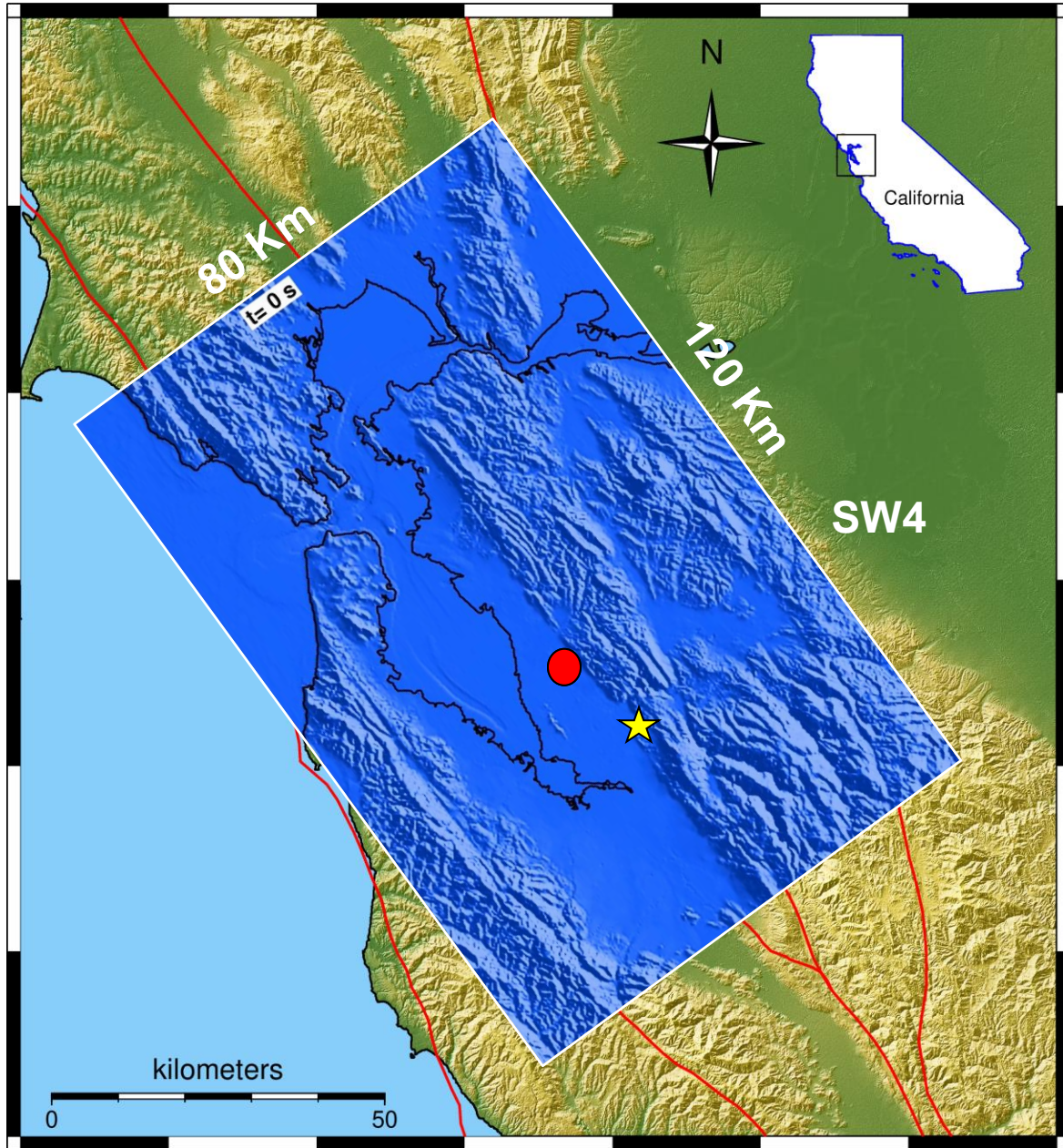
- **Starting point:** CPU-only code written in C by physicists. Code had complex nested loops (fourth-order finite difference stencil) that the compiler struggled to optimize, find SIMD.
- **ECP accomplishments**
 - Algorithmic improvements using curvilinear mesh refinement improved scientific work-rate by a factor of 2.85
 - Rewrote code in C++ with RAJA, with ZFP data compression to save sufficient data to maintain adequate precision in stored data
 - Infrastructure simulations now include strong coupling with OpenSees soil/building modeling and using in soil-structure interaction models; help gain insight into areas of maximum risk.



Achieved a **3500X improvement in computational performance** compared to initial baseline on Cori (~30 PF KNL system); Simulation of regional-scale ground motions at frequencies of engineering interest (5-10 Hz) now within reach.

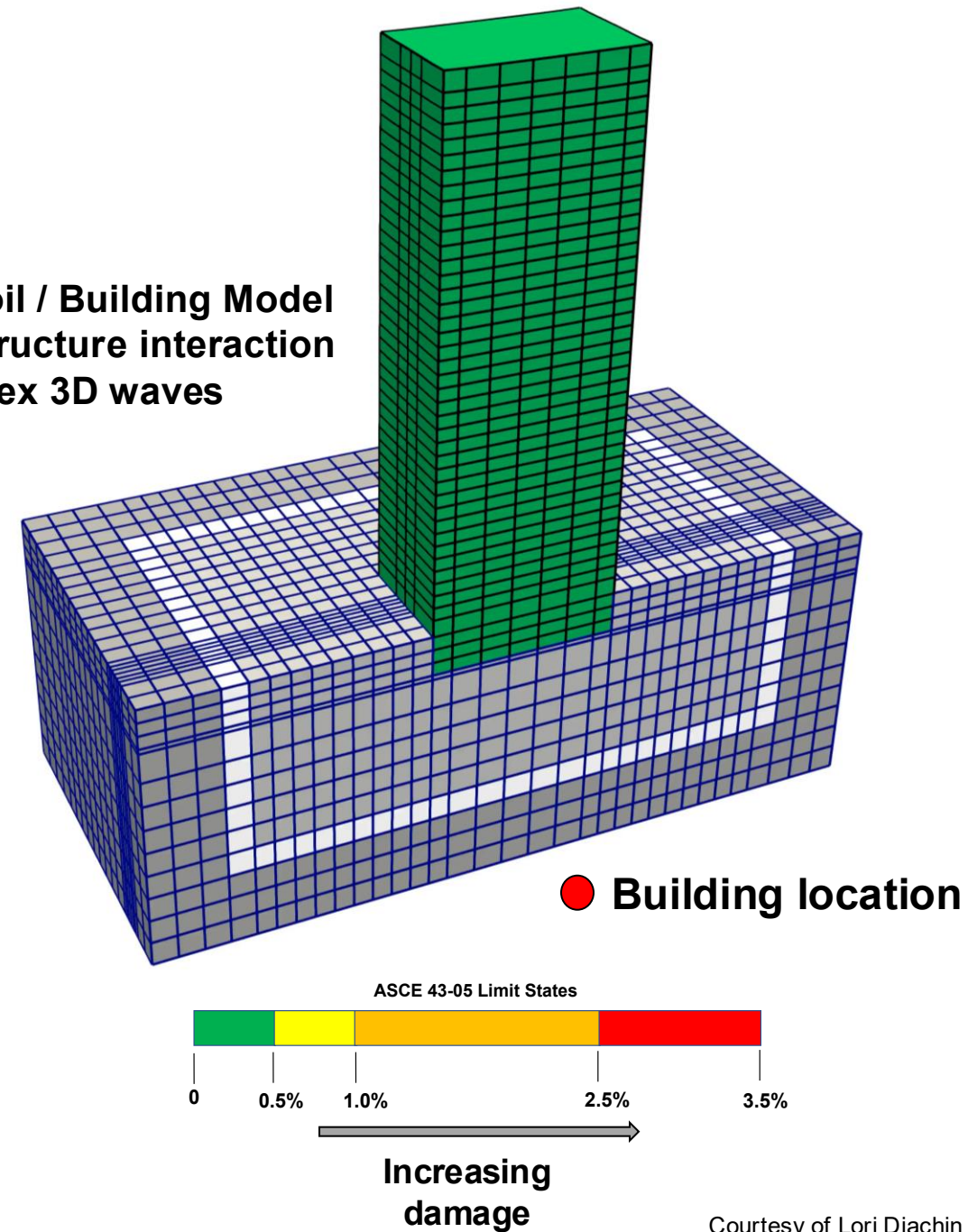


Regional Geophysics Model Fmax 10 Hz V_{min} 140 m/s
(M7 Hayward fault earthquake) 391 Billion grid points



Local Soil / Building Model

- Soil-structure interaction
- Complex 3D waves



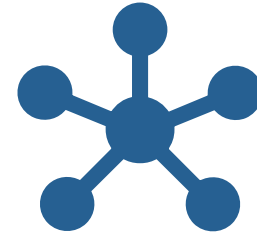
There where several key ingredients required for a successful ECP Application Development Project



Algorithmic innovation



Porting

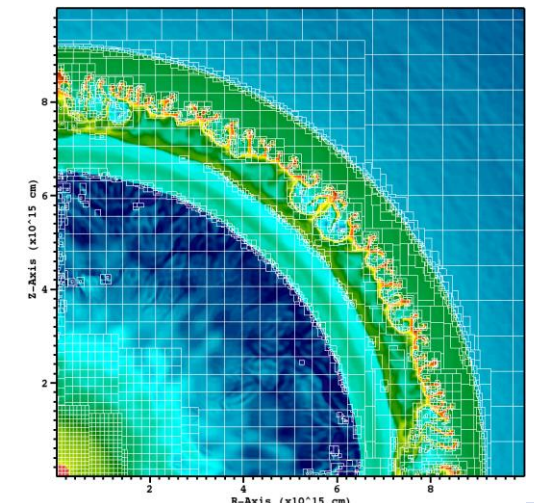
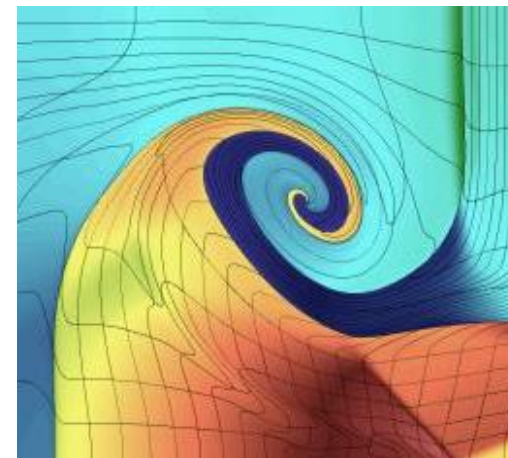
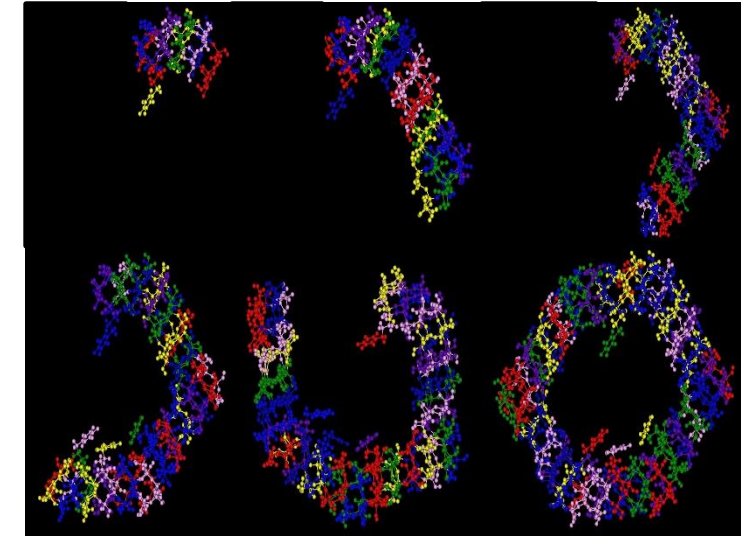


Integration

Algorithmic innovation: domain-driven adaptations critical for making efficient use of exascale systems

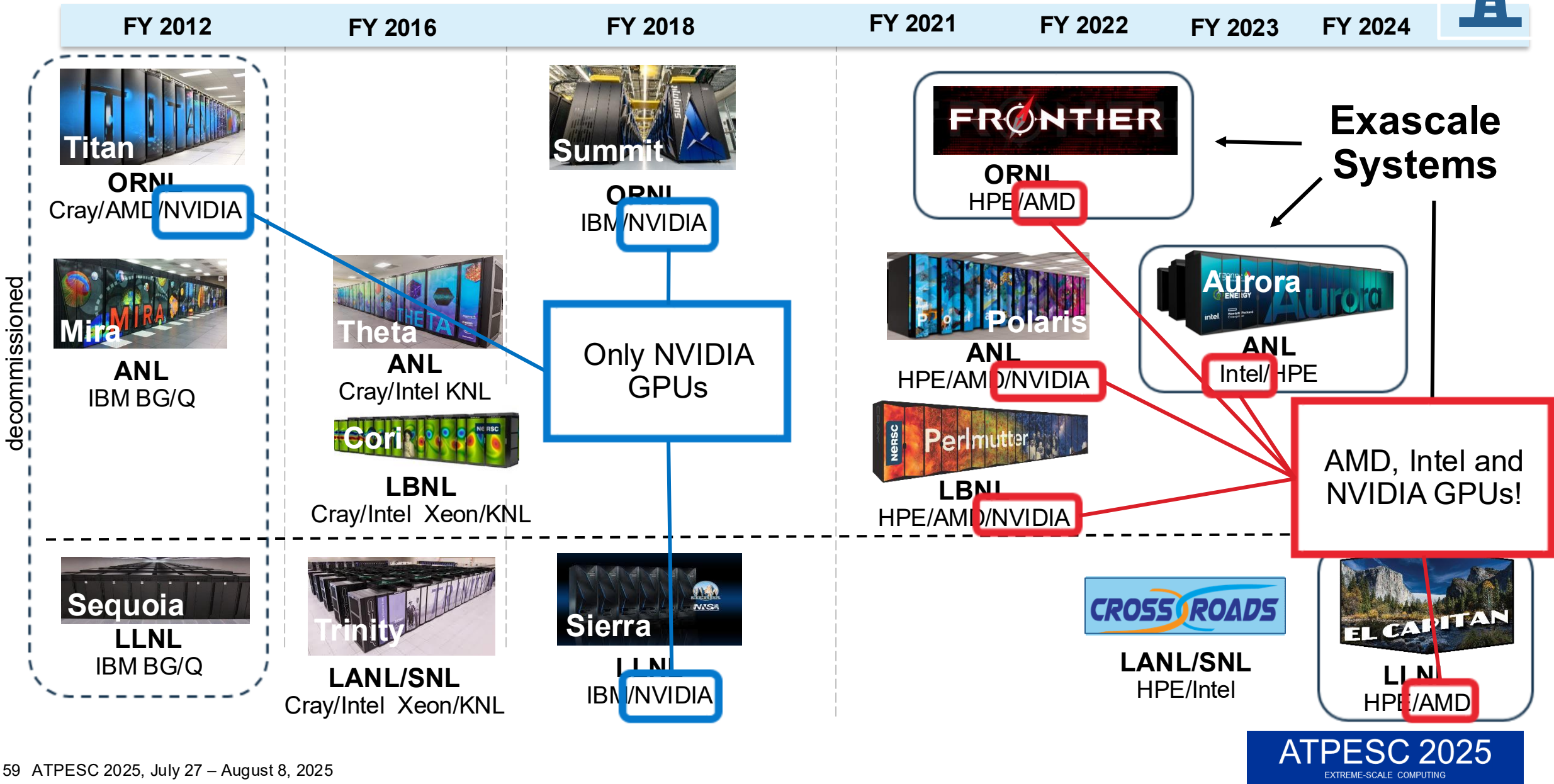


- Inherent strong scaling challenges on GPU-based systems
 - Ensembles vs. time averaging
 - Fluid dynamics, seismology, molecular dynamics, time-stepping
- Increase dimensions of (fine-grained) parallelism to feed GPUs
 - Ray tracing, Markov Chain Monte Carlo, fragmentation methods
- Localized physics models to maximize "free flops"
 - MMF, electron subcycling, enhanced subgrid models, high-order discretizations
- Alternatives to sparse linear systems
 - Higher order methods, Monte Carlo
- Reduced branching
 - Event-based models

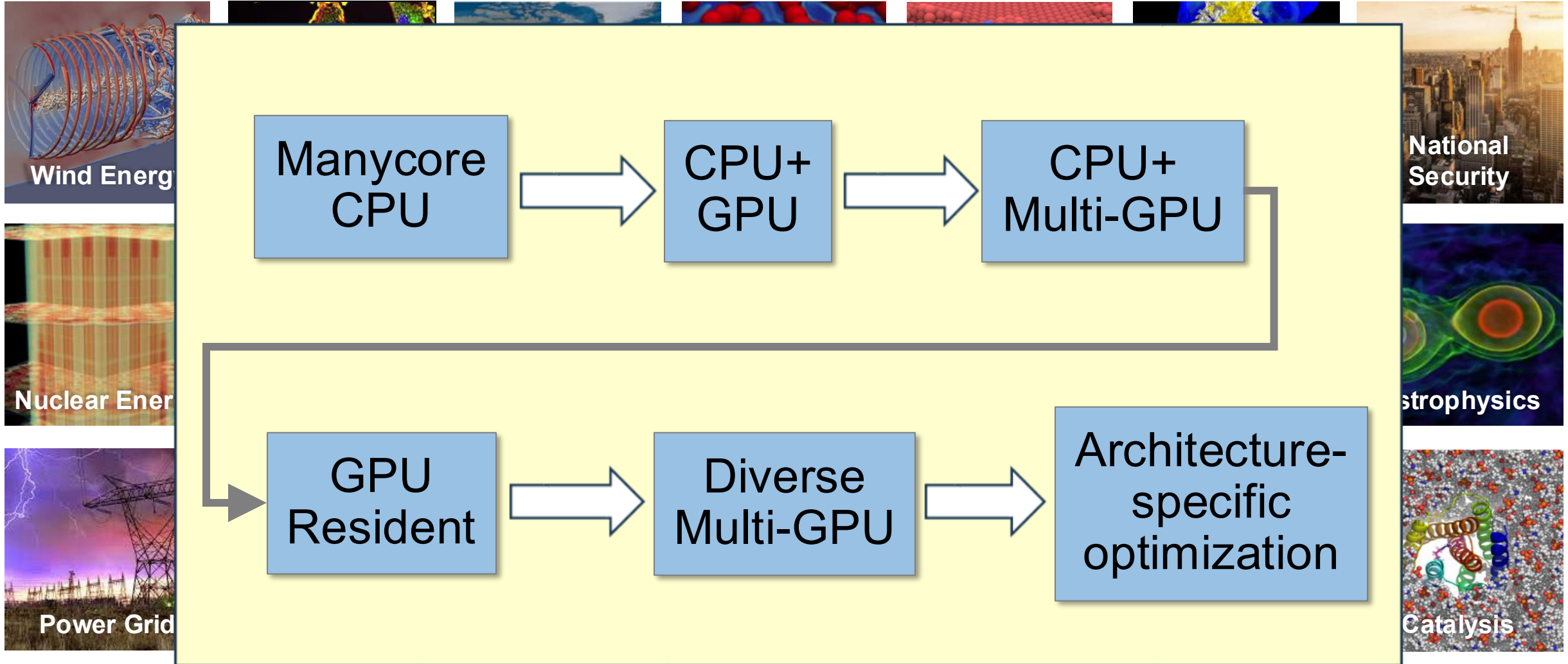


ATPESC 2025
EXTREME-SCALE COMPUTING

HPC systems have come a long way since ECP's inception



Exascale applications were designed to be flexible and adaptive



ECP applications teams used several different programming models to achieve performance portability



GPU-specific kernels

- Isolate the computationally-intensive parts of the code into CUDA/HIP/SYCL kernels.
- Refactoring the code to work well with the GPU is the majority of effort.

Loop pragma models

- Offload loops to GPU with OpenMP or OpenACC.
- Most common portability strategy for Fortran codes.

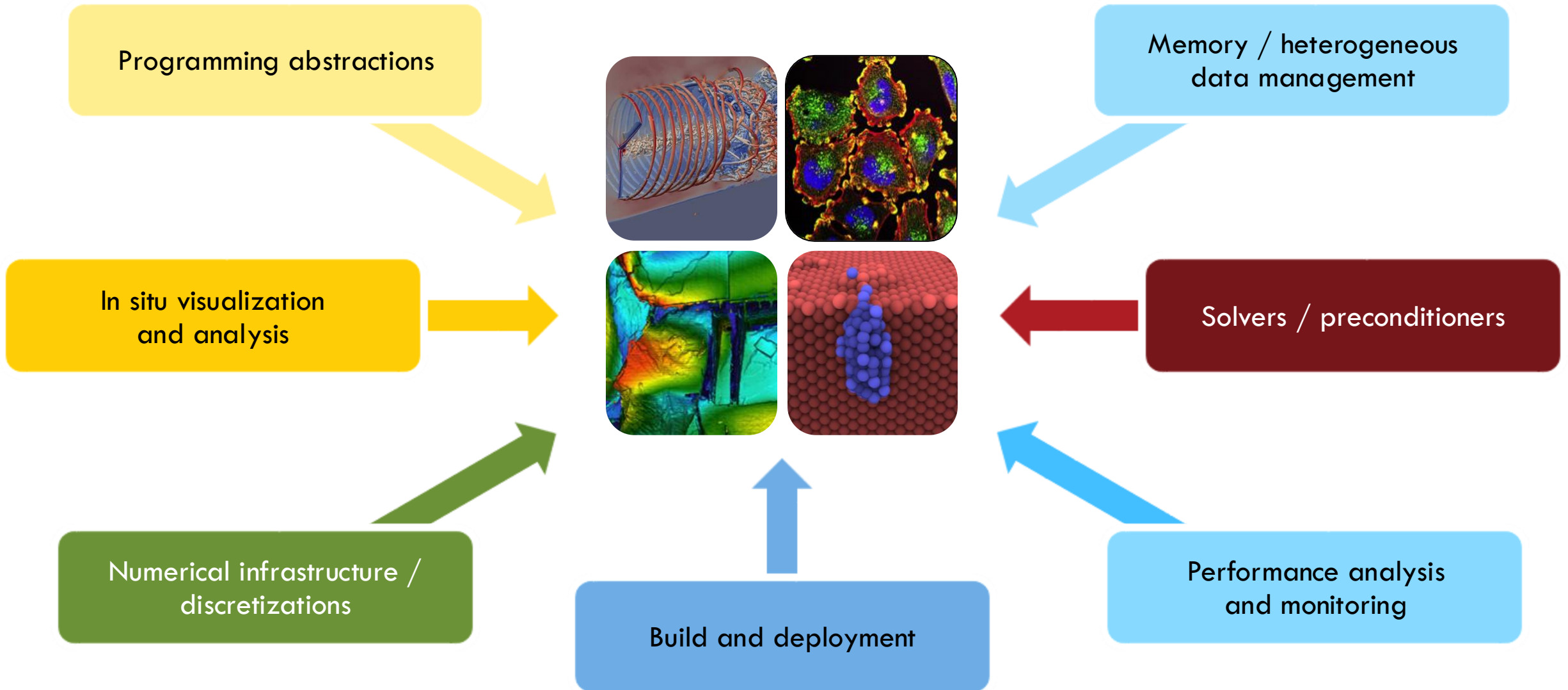
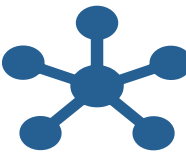
C++ abstractions

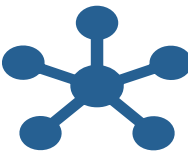
- Fully abstract loop execution and data management using advanced C++ features.
- Kokkos and RAJA developed by NNSA in response to increasing hardware diversity.

Co-design frameworks

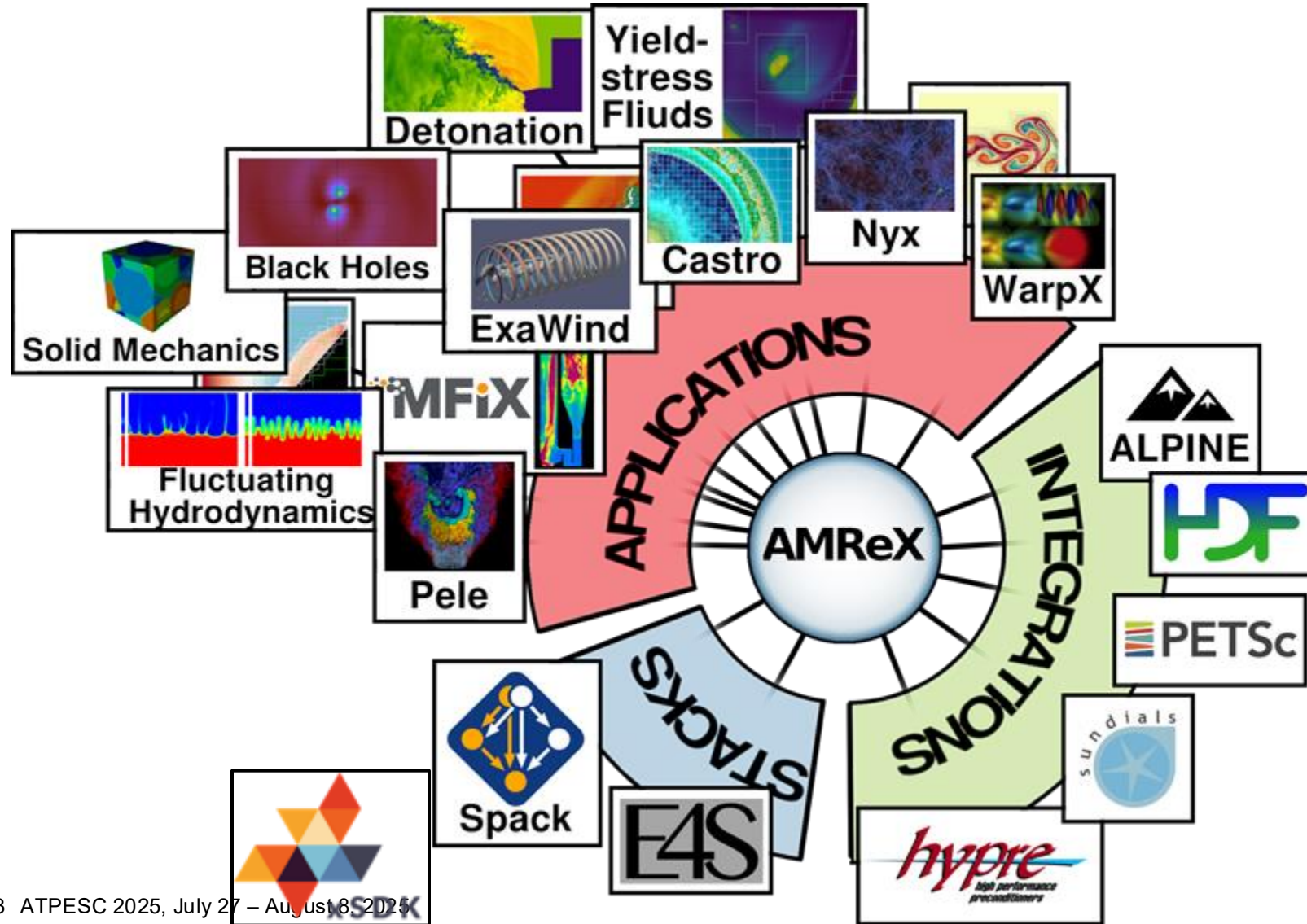
- Design application with a specific motif to use common software components
- Depend on co-design code (e.g. CEED, AMReX) to implement key functions on GPU.

The success of exascale applications relied on significant software infrastructure





Integration of multiple technologies allow users of higher-level frameworks, e.g., AMReX, access many different technologies



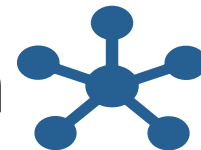
Software Integrations:

- SUNDIALS – Chemical reactions, time integrators
- *hypre*, PETSc – Linear solvers on mesh data
- In-situ: Ascent, Sensei
- Offline visualization: VisIt, Paraview, yt
- IO: HDF5, ADIOS

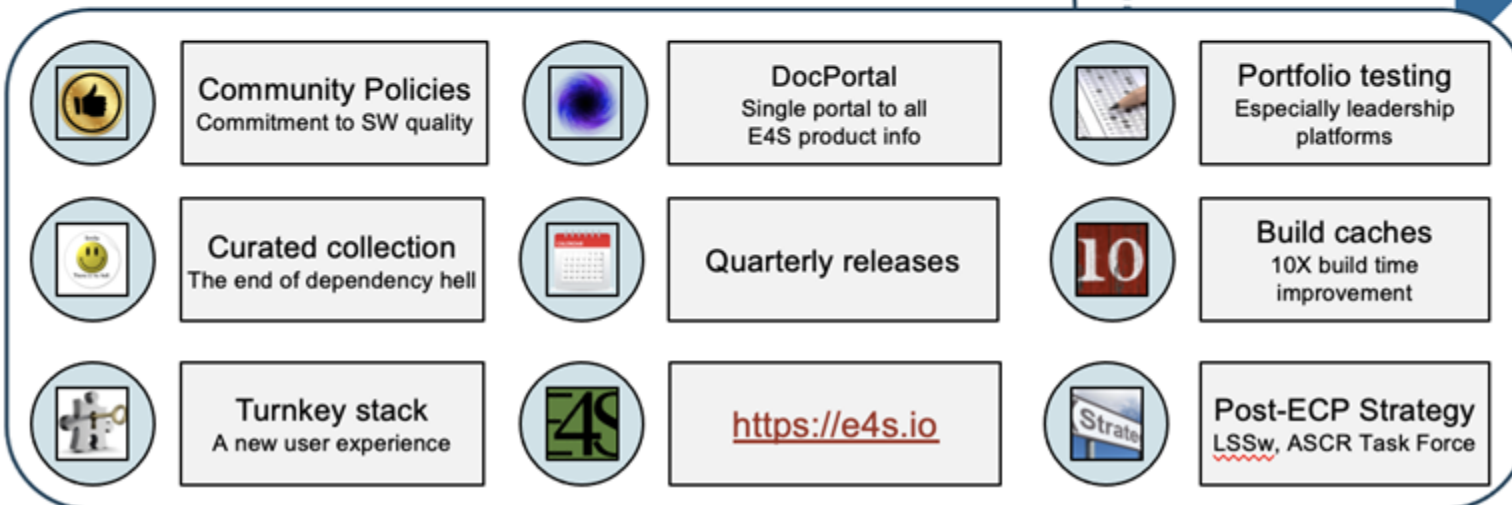
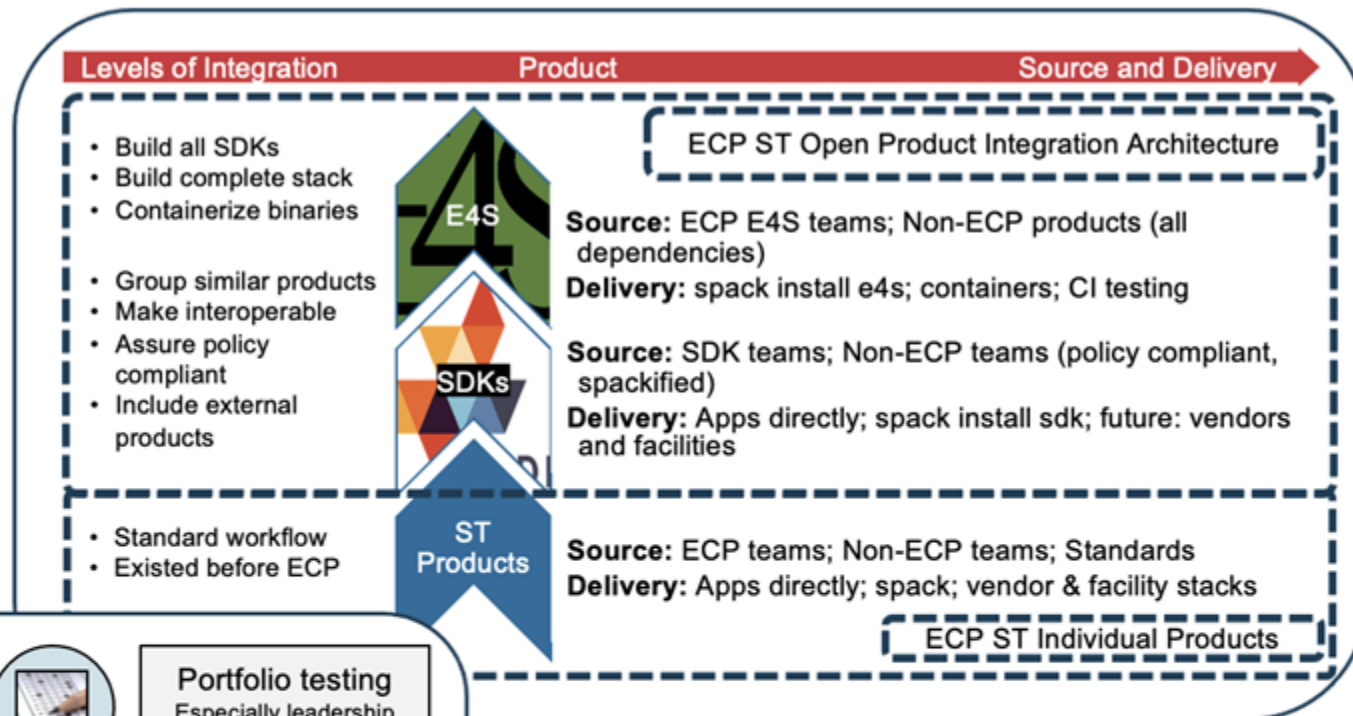
Software Stacks:

- Spack
 - Smoke test – CUDA, AMD HIP
- xSDK
- E4S

ECP software technologies were broadly deployed through the Extreme Scale Scientific Software Stack (E4S)



- ECP Software Technology lead: Mike Heroux (SNL)
- E4S: HPC software ecosystem – a curated software portfolio
- A **Spack-based** distribution of software tested for interoperability and portability to multiple architectures
- Available from **source, containers, cloud, binary caches**
- Not a commercial product – an open resource for all
- Supported by DOE and commercial entities (Paratools)
- Growing functionality: November 2023: E4S 23.11 – 100+ full release products



<https://e4s.io>

E4S lead: Sameer Shende (U Oregon)

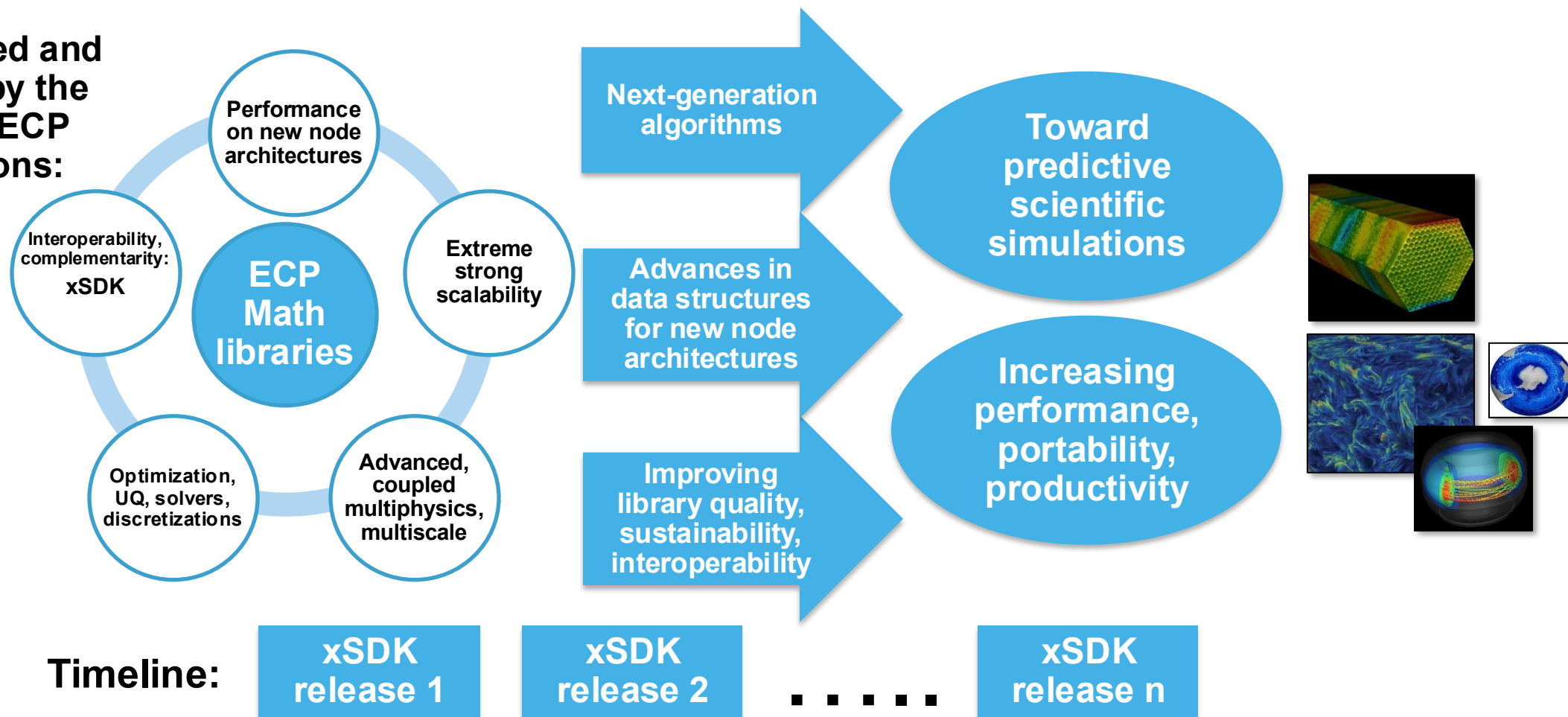


<https://spack.io>

Spack lead: Todd Gamblin (LLNL)

xSDK: Primary delivery mechanism for ECP math libraries' continual advancements toward predictive science

As motivated and validated by the needs of ECP applications:



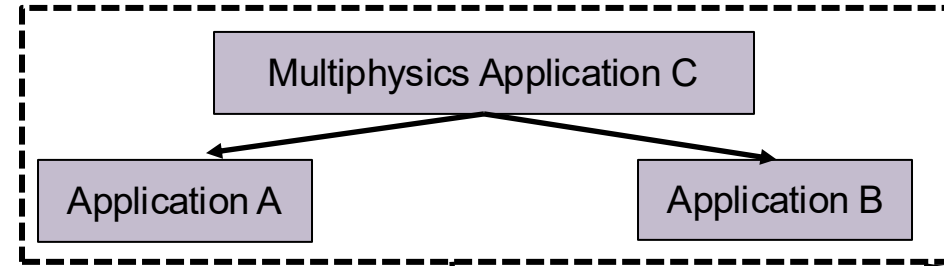


xSDK Version 1.1.0: November 2024

<https://xsdk.info>

<https://xsdk.info>

Each xSDK member package uses or can be used with one or more xSDK packages, and the connecting interface is regularly tested for regressions.



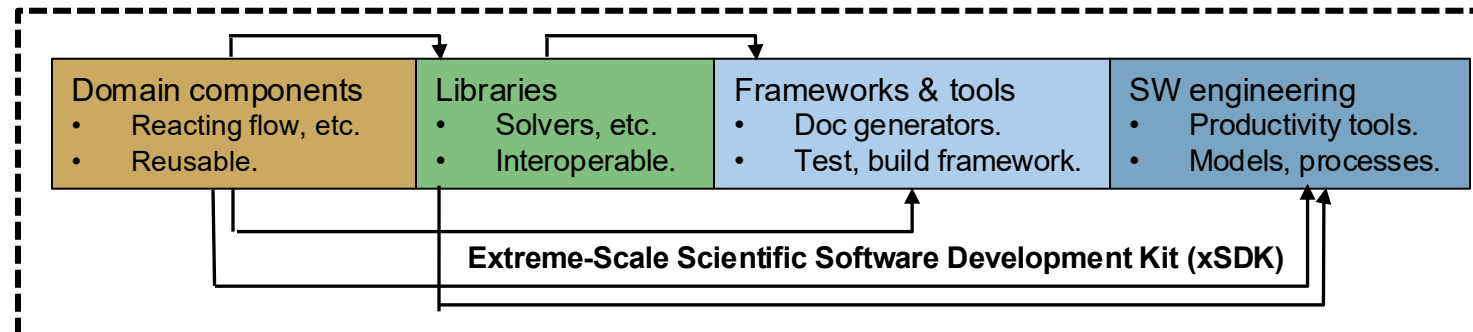
xSDK functionality, Nov 2024

Tested on Linux work stations



November 2024

- 25 math libraries
- 17 mandatory xSDK community policies
- Spack xSDK installer

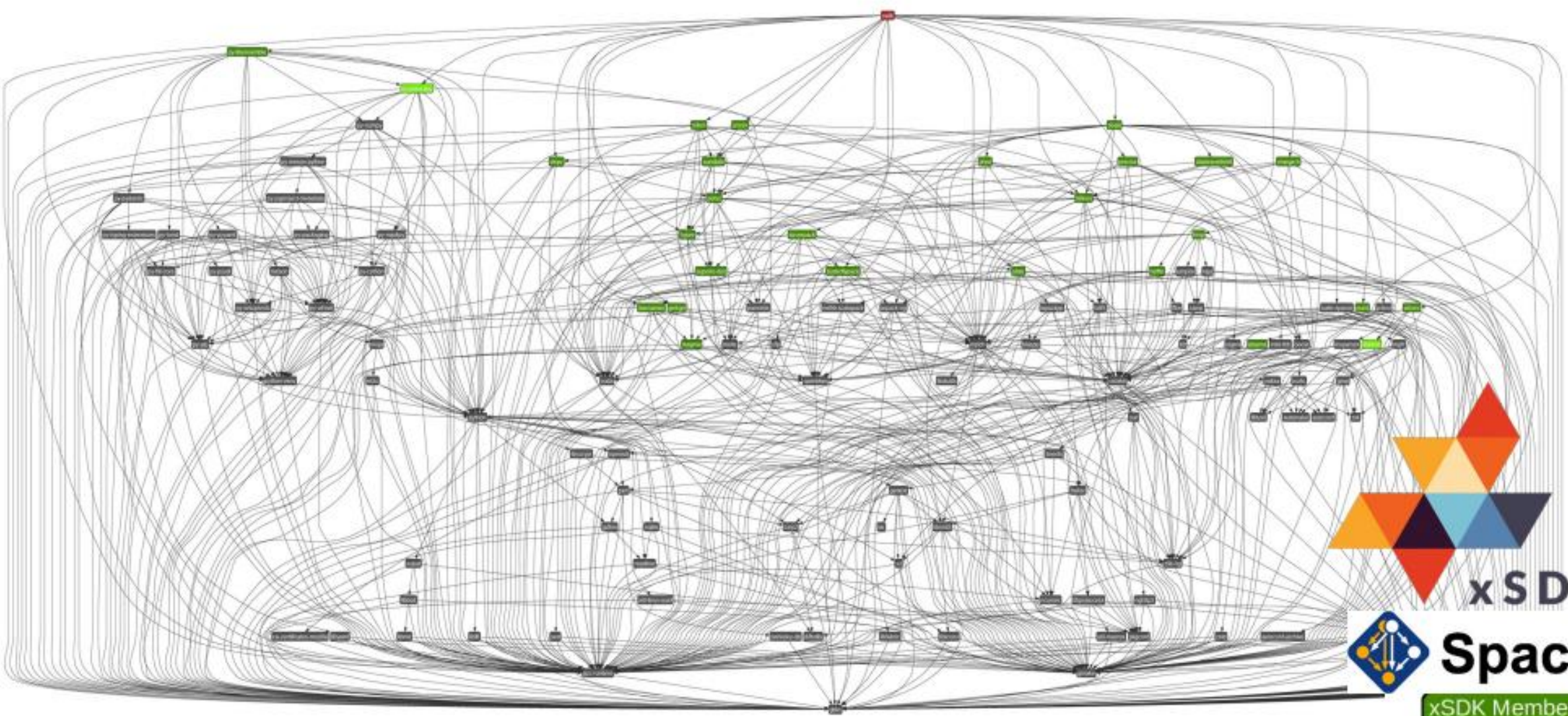


Impact: Improved code quality, usability, access, sustainability

Foundation for work on performance portability, deeper levels of package interoperability

ATPESC 2025

EXTREME-SCALE COMPUTING





xSDK: <https://xsdk.info>

Building the foundation of an extreme-scale scientific software ecosystem

xSDK community policies: Help address challenges in interoperability and sustainability of software developed by diverse groups at different institutions

<https://github.com/xsdk-project/xsdk-community-policies>

xSDK compatible package: must satisfy the mandatory xSDK policies (M1, ..., M17)

Topics include configuring, installing, testing, MPI usage, portability, contact and version information, open-source licensing, namespacing, and repository access

Also specify **recommended policies**, which currently are encouraged but not required (R1, ..., R8)

Topics include public repository access, error handling, freeing system resources, and library dependencies, [documentation quality](#)

xSDK member package:

- (1) Must be an xSDK-compatible package, *and*
- (2) it uses or can be used by another package in the xSDK, and the connecting interface is regularly tested for regressions.

xSDK policies 1.0.0: Feb 2023

- Facilitate combined use of independently developed packages

Impact:

- Improved code quality, usability, access, sustainability
- Foundation for work on deeper levels of interoperability and performance portability

We encourage feedback and contributions!

The Exascale Computing Project has delivered on its mission needs

Deliver a long-term, **sustainable software ecosystem** that can be used and maintained for years to come

- ❖ E4S deployed at HPC facilities around the US and the world
- ❖ 76 HPC products available for computing at all scales
- ❖ Performance portability tools developed and widely used

Promote the **health of the US HPC industry**

- ❖ Six vendors funded under PathForward; outcomes realized in exascale systems
- ❖ Accelerator-based computing lowers cost of energy across the board
- ❖ The ECP Industry and Agency Council stimulates consumption of HPC resources

Ensure that exascale systems can be used to deliver **mission-critical applications**

- ❖ ECP applications demonstrate outstanding performance and capabilities at exascale
- ❖ Previously unattainable results in real-world challenge problems
- ❖ ECP lessons learned pave the way for many additional applications to leverage accelerator-based computing

Maintain **international leadership in HPC**

- ❖ Frontier is the world's first exascale machine – in part due to ECP/ECI investments
- ❖ Aurora is the world's second exascale system
- ❖ 1000+ researchers trained in GPU computing

HandsOn Lessons

- Structured meshing & discretization
- Unstructured meshing & discretization
- Krylov solvers & preconditioners
- Sparse direct solvers
- Nonlinear solvers
- Time integration
- Numerical optimization



ATPESC 2025 Hands On Lessons

Meshing and Discretization with AMReX	A Block Structured Adaptive Mesh Refinement Framework
Unstructured Meshing & Discretization with MFEM	Finite Elements and Convergence
Krylov Solvers and Algebraic Multigrid with hypre	Demonstrate utility of multigrid
Iterative Solvers & Algebraic Multigrid (with Trilinos, Belos & MueLu)	Introduction to Krylov Solvers and Preconditioning, with emphasis on Multigrid
Sparse, Direct Solvers with SuperLU	Role and Use of Direct Solvers in Ill-Conditioned Problems
Rank Structured Solvers with STRUMPACK	Using STRUMPACK for dense and sparse linear systems
Nonlinear Solvers with PETSc	Introduction to Nonlinear Solvers: Newton-Krylov Methods and Nonlinear Preconditioning

And more ...

Github pages site:

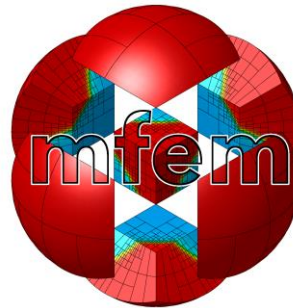
<https://xsdk-project.github.io/MathPackagesTraining2025/lessons/>

If you haven't yet done so, please choose which session you plan to attend!

Time	Room?	Room?
8:30 – 9:30	Introduction to Numerical Software – Ulrike Yang	
9:30 – 10:45	Structured Discretization (AMReX) – Andrew Myers, Weiqun Zhang	Unstructured Discretization (MFEM/PUMI) – Mark Shephard, Cameron Smith, Mark Stowell
10:45 – 11:15	Break, Subject Matter Expert (SME) Selections, Panel Questions	
11:15 – 12:30	Iterative Solvers & Algebraic Multigrid (hypre) – Daniel Osei-Kuffuor, Ulrike Yang	Direct Solvers (SuperLU, STRUMPACK) – Sherry Li, Yang Liu
12:30 – 1:30	Lunch, SME Selections, Panel Questions	
1:30 – 2:45	Nonlinear Solvers (PETSc) – Richard Mills	Time Integration (SUNDIALS) – David Gardner
2:45 – 3:15	Break, SME Selections, Panel Questions Due	
3:15 – 4:30	Optimization (TAO) – Toby Isaac	Iterative Solvers & Algebraic Multigrid (Trilinos/ Belos/MueLU) – Christian Glusa, Graham Harper
4:30 – 5:15	Panel: Extreme-Scale Numerical Algorithms and Software	
5:15 – 6:30	Optional Activity: SME Speed-dating	
6:30 – 7:30	Dinner	
7:30 – 9:00	After-Dinner Talk: Jim Demmel	

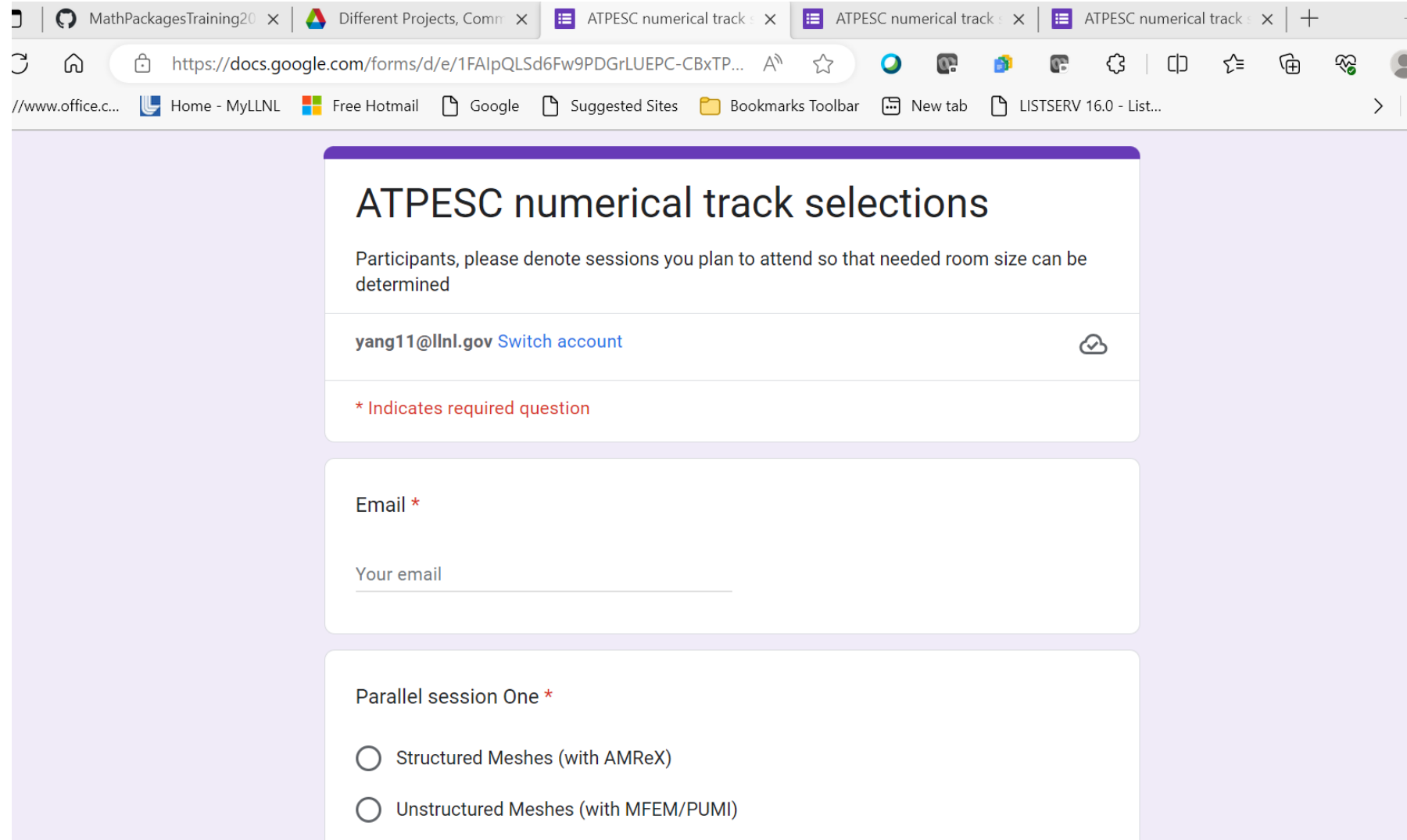
Room Choice

- Please raise your hand if you want to attend
 - Structured meshing
 - Unstructured meshing



Choose which lecture you want to attend!

ATPESC numerical track selections



The screenshot shows a web browser with multiple tabs open, including 'MathPackagesTraining20', 'Different Projects, Comm...', and several 'ATPESC numerical track' tabs. The active tab displays a Google Form titled 'ATPESC numerical track selections'. The form's purpose is to gather session preferences for room size determination. It includes a user identification section with the email 'yang11@llnl.gov' and a 'Switch account' link. A red asterisk indicates required questions. The first required question is 'Email', with a text input field labeled 'Your email'. The second required question is 'Parallel session One', which has two radio button options: 'Structured Meshes (with AMReX)' and 'Unstructured Meshes (with MFEM/PUMI)'.

ATPESC numerical track selections

Participants, please denote sessions you plan to attend so that needed room size can be determined

yang11@llnl.gov [Switch account](#)

* Indicates required question

Email *

Your email

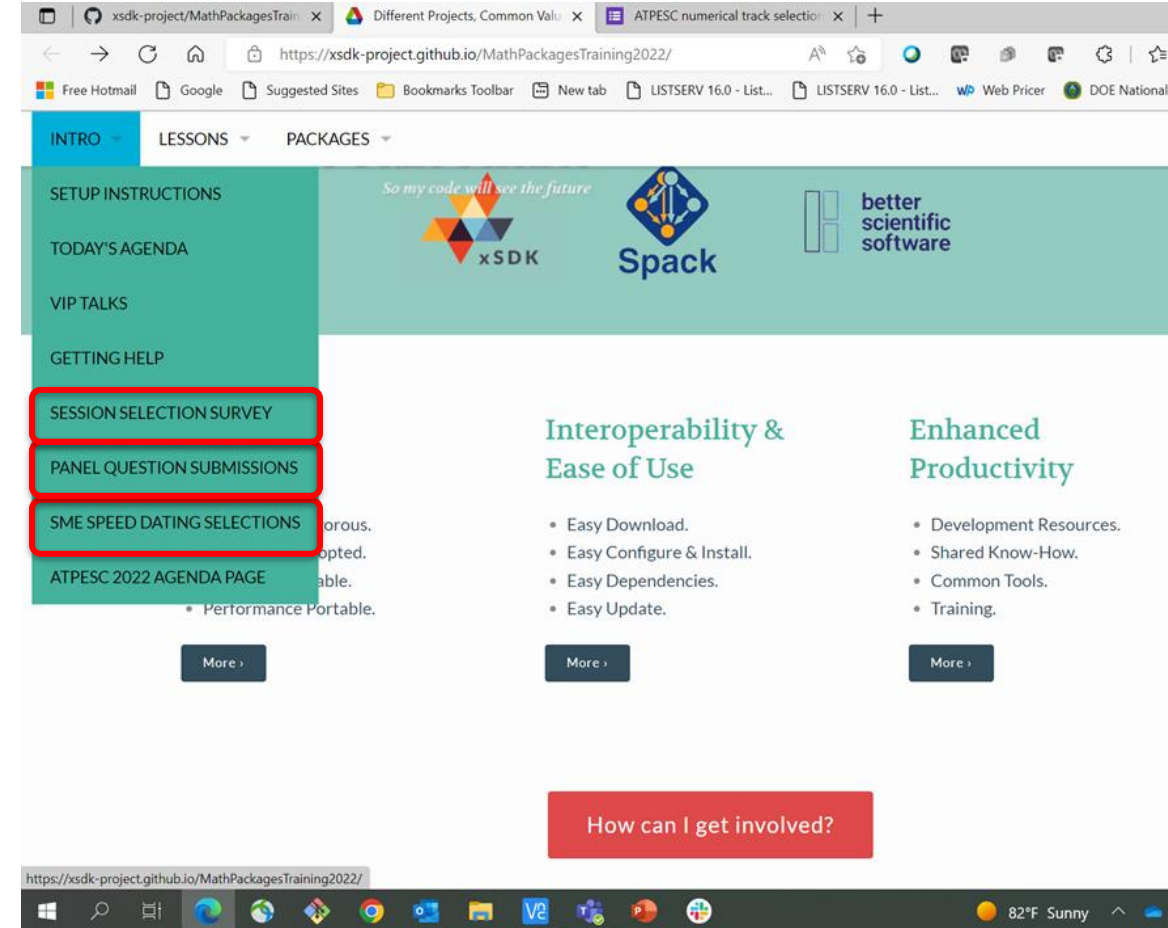
Parallel session One *

☐ Structured Meshes (with AMReX)

☐ Unstructured Meshes (with MFEM/PUMI)

Next steps

- If you haven't done so
 - Choose which session you will attend!
- During breaks and lunch
 - Submit questions for panelists (optional)
 - Sign up for discussions with numerical software developers (optional)
 - [Your email address](#)
 - Complete by 4:30 pm CDT



Thank you to all ATPESC staff



Special thanks to Ray Loy
and
Kathy Gorgan, Yasaman Ghadar,
Haritha Siddabathuni Som,
Johnny Marquez and ...

**For their outstanding work in
running the ATPESC program**

**And thank you to all ATPESC attendees for
engaging questions and discussions!**



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